Renewable and Sustainable Energy Reviews 151 (2021) 111567

Contents lists available at ScienceDirect

Renewable and Sustainable Energy Reviews

journal homepage: www.elsevier.com/locate/rser

Electric vehicle routing problem: A systematic review and a new comprehensive model with nonlinear energy recharging and consumption

Yiyong Xiao [a,1], Yue Zhang [a,1], Ikou Kaku [b], Rui Kang [a,*], Xing Pan [a,*]

[a] School of Reliability and Systems Engineering, Beihang University, Beijing, 100191, China
[b] Department of Environmental Management, Tokyo City University, Yokohama, 224-8551, Japan

ARTICLE INFO

Keywords:
Electric vehicle routing problem
Emission
Mathematical programming
Solomon's instances

ABSTRACT

Electric vehicles (EVs) are witnessing a surge in demand and production for their environmental benefits. This study reviews the development of electric vehicle routing problem (EVRP) in transport logistics. The review found that few existing EVRP models considered integrally the unique characteristics of EVs, such as the nonlinear charging function, nonlinear electricity-consumption function, and dynamic driving ranges. This study develops a new comprehensive model of the EVRP that considers a general energy/electricity consumption function for EVs that factor in energy losses (aerodynamic, tire friction, drivetrain, ancillary, and kinetic/potential losses), nonlinear charging function with the piecewise linearization technique, efficient visits to charge stations, and continuous decision variables for speed, payload, travel time, recharging, etc. The newly developed model is much closer to reality by integrating most of the unique characteristics of EVs known so far, and therefore it advances the state-of-the-art in EVRP. Computational experiments on Solomon's instances were done to exhibit the effectiveness and efficiency of the developed model under the effects of air conditioner, battery capacity, and travel speed. These experiments demonstrated that the model can obtain more practical logistics plans for an EV fleet with a lower total emissions cost and higher energy utilization.

1. Introduction

In the 21st century, the fast increase of green-house gas (GHG) emissions leads to a continuous rise in the global average temperatures (McCarthy et al., 2010) [1]. The transportation sector has contributed more than 23 % of the global GHG emissions because it consumes a large share of the global oil production (Sims et al., 2014) [2]. EVs present a promising opportunity to improve climatic conditions owing to their eco-friendly characteristics such as low GHG emissions, higher energy efficiency, and lower noise pollution (Zuo et al., 2019; Soleimani et al., 2018; Desaulniers et al., 2016) [3–5]. The equivalent GHG emissions of EVs can be reduced to one-fifth of the original value by fossil-fuel-powered vehicles (Li et al., 2015) [6], and the cost of electricity consumption can be reduced to maximally one-tenth of the original value (Xiao *et al.*, 2019) [7]. Accordingly, to relieve the stress of environmental degradation and climate change (Qian et al., 2011) [8], electric vehicles (EVs) is recommended as the technological measure toward the reduction of GHGs (Mashayekh et al., 2012; Eberle and Helmolt, 2010; Offer et al., 2010; Pucher and Buehler, 2008) [9–12].

Although obstacles including limited driving range and long charging time still hamper the promotion of EVs, global sale of the EVs is increasing steadily, especially in the Chinese market, where it has been growing at an annual rate of more than 50 % in recent years (IEA, 2018) [13]. The electric vehicle routing problem (EVRP) has been becoming a hot research topic of modern logistics that the academia and industry have been attempting to explore and solve. According to the characteristics and application scenarios of EVs, many EVRP models have been proposed for various fields. For instance, Erdoğan and Miller-Hooks (2012) [14] formulated the green vehicle routing problem (G-VRP) and considered charging station (CS) visits as the variable on the basis of the traditional vehicle routing problem (VRP). Goeke and Schneider (2015) [15] developed a nonlinear energy/electricity consumption rate (ECR) model for EVs. Felipe et al. (2014) [16] and Montoya et al. (2017) [17] took charge time as a decision variable into the EVRP model.

Although several studies focus on EVRP, none, to the best of the authors' knowledge, provide a comprehensive review or classification of EVRP models. In this study, a comprehensive and systematic review is

* Corresponding authors. School of Reliability and Systems Engineering, Beihang University, 37# Xueyuan Road, Haidian District, Beijing, 100191, China.
E-mail addresses: xiaoyiyong@buaa.edu.cn (Y. Xiao), zhangyue1127@buaa.edu.cn (Y. Zhang), kakuikou@tcu.ac.jp (I. Kaku), kangrui@buaa.edu.cn (R. Kang), panxing@buaa.edu.cn (X. Pan).
[1] These two authors contributed equally to this work

https://doi.org/10.1016/j.rser.2021.111567
Received 4 October 2020; Received in revised form 11 May 2021; Accepted 4 August 2021
Available online 14 August 2021
1364-0321/© 2021 Elsevier Ltd. All rights reserved.

Abbreviations

A/C	Air conditioning
AFS	Alternative fueling station
CS	Charging station
C–W	Clarke and Wright
ECR	Energy/electricity consumption rate
EV	Electric vehicle
EVRP-C	EVRP with charging
EVRP-TW	EVRP with time window
EVRP-MF	EVRP with mixed fleet
EVRP-MFTW	EVRP with mixed fleet and time windows
EVRP	electric vehicle routing problem
G-VRP	Green vehicle routing problem
GA	Genetic algorithm
GHG	Green-house gas
LCF	Linear charging function
NCF	Nonlinear charging function
MILP	Mixed-integer linear programming
MD-EVRP-I	multi-depot EVRP with inter depot routes
SOC	State of charge
SVR	speed-varying travel range
VRP	vehicle routing problem

provided to summarize various modeling problems in the development of EVRP, including the charging process of EVs, CS visits, and electricity/energy consumption models. In addition, by considering the actual application types of vehicle routing problems, this review introduces the main developments in EVRP with charging (EVRP-C), EVRP with time window (EVRP-TW), and EVRP with mixed fleet (EVRP-MF), and then categorizes various methods for solving them. However, it was found that EVRP models proposed in the literature do not consider all types of practical factors simultaneously, such as the partial charging policy with nonlinear charging function and flexible charging time, multiple visits of CSs, and dynamic driving range with nonlinear ECR function composed of actual end uses. To consider various practical factors and produce optimized solutions, this study developed a new comprehensive model to advance the state-of-the-art in EVRP. Three major contributions were made to existing research: (1) A taxonomic literature review of the EVRP, including models containing the problems, actual application types, and approaches to solving the EVRP; (2)A physical model of the ECR for one unit of traveled distance established by breaking down the different end-uses of consumed battery capacity. This improvement to the ECR model allows it to use the dynamic range of EVs in a more practical manner; (3)A new comprehensive model for EVRP considering the electricity/energy consumption model and nonlinear charge function based on the mixed-integer linear programming (MILP).

The rest of the article is organized as follows. In Section 2, a systematical review of the EVRP is summarized. In Section 3, a dynamic ECR model with speed and load as the decision variables is established, and methods to linearize the quadratic function and inverse proportional function are introduced. In Section 4, new models of CS visit and battery charging are formulated. In Section 5, a new continuous model for EVRP based on MILP is proposed. Computational experiments are done on existing Solomon's instances with result analysis. Finally, Section 7 presents conclusions.

2. Literature review

In sub-Sections 2.1-2.3, various modeling problems of EVRP are summarized, including CS visit, charge time optimization, and electricity/energy consumption. In Section 2.4, the developments of EVRP-C, EVRP-TW and EVRP-MF are developed. In Section 2.5, various solutions to solve the EVRP models are categorized. Finally, a review summary is provided in Section 2.6 to conclude some remaining challenges. The general framework of the EVRP review is shown in Fig. 1.

2.1. Charge station visit model

Conrad and Figliozzi (2011) [18] incorporated the visits of recharging/refueling station in the vehicle routing problem, while assuming that the recharging/refueling services were available at only a few of the customer places. Erdoğan and Miller-Hooks (2012) [14] treated the recharging/refueling station, namely, alternative fueling station (AFS) in their G-VRP model, as special nodes under the framework of vehicle routing problem. Since then, CS visit has become a major study area within the EVRP. However, it poses a challenge to the EVRP model owing to the following four characteristics of CS nodes. First, it is not necessary for vehicles to visit each CS node, even though all customer nodes should be served. Second, each CS node can be visited by multiple vehicles unlimited times, while each customer node can only be visited once by one vehicle. Thirdly, each CS is limited by capacity. A long charge time indicates high occupancy at the CS, and thus advanced reservations are required sometimes. Fourthly, there may be a large number of CS nodes, much more than the customer nodes, which reduces the efficiency of the model solution.

Accordingly, many EVRP studies have focused on modeling the CS visit. Erdogan and Miller-Hooks (2012) [14] duplicated the AFS visit into multiple dummy nodes, so that the EVs could visit an AFS once or more than once. This technique of dummy vertices was introduced by Bard et al. (1998) [19] for their application of reloading vehicles for delivery. Based on this technique, several researchers have proposed other variants of the EVRP (Felip et al., 2014; Goeke and Schneider, 2015; Yang and Sun, 2015; Hiermann et al., 2016; Montoya et al., 2017; Schiffer and Walther, 2017) [15–17,20,21,39].

However, disadvantages of using dummy nodes were also pointed out by Koc and Karaoglan (2016) [23]. Because the maximum times of visits to a CS cannot be exactly estimated, so an CS must be duplicated as many times as possible to guarantee the optimal solution is retained, which could increase inevitably and considerably the computational complexity. In addition, an arc detour formulation is also added to EVRP models, by which a three-dimension binary decision variable, x_{ijc}, is introduced to indicate if an EV would visit a CS c en route the journey from customer i to customer j. Leggieri and Haouari (2017) [24] improved this formulation by considering the duration limit. However, the three-dimension decision variable has a large combinational space when the CS number is large, which increases the formulation complexity to a level of $O\ (|N|^2 \times |C|)$, where $|N|$ represent the customer/node number and $|C|$ represent the CS number. Zuo et al. (2019) [3] simplified the formulations of arc detour by using the traditional two-dimension variable associated with another two-dimension variable indicating whether an EV will visit CS. Therefore, this technique is more efficient and has the same formulation complexity as the traditional VRP.

In this study, a new fixed arc detour technique is developed. The difference between this method and the existing method is on the complexity. In this method, each arc (from one customer to another) is supposed to have a fixed optimal CS visit detour that can be pre-defined early. Thus, complex mathematical models are no longer required to determine the occurrence of a CS visit, and only one binary variable is needed to indicate whether the nearest CS of the arc is visited or not when traversing the arc. Table 1 summarizes and compares the existing techniques of CS visit occurrence determination.

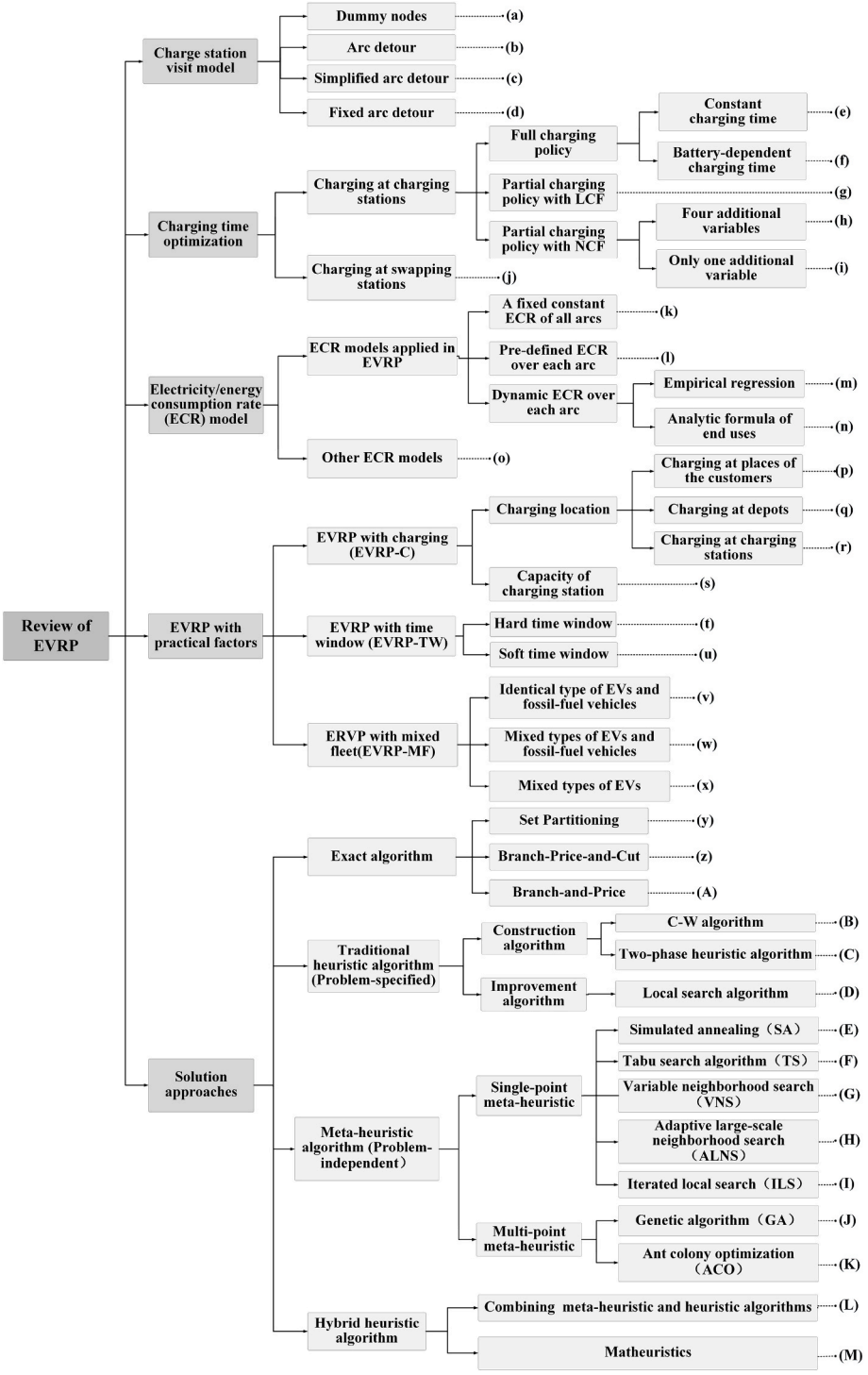


Fig. 1. Framework of EVRP review.

2.2. Charging time optimization in EVRP

For practical purposes, it is important to determine the optimal charging time of an EV in the EVRP during a CS visit, because, at present, EVs take a longer time to fully recharge than conventional fossil-fuel-powered vehicles. In the actual situation, the charging function is nonlinear owing to changes in terminal voltage and current during the charging process. The state of charge (SOC) increases linearly with time at first, and then increases concavely with time until full charge (Pelletier et al., 2017) [26]. The typical charging curve of the SOC over time as plotted by Hõimoja et al. (2012) [27].

Accordingly, characteristics of the charging process should be considered into the EVRP modeling. In existing models, methods of charging-time optimization can be classified into full charging policy, partial charging policy with linear charging function (LCF), partial charging policy with nonlinear charging function (NCF), and battery swapping policy. Fig. 2 summarizes the framework of charging-time optimization in the EVRP and the corresponding literature.

Regarding full charging policy, each time an EV visits a CS, its battery should be fully charged. Researchers studying full charging policy

Table 1
Comparison of four techniques of CS visit.

CS visit method	Method description	Advantages	Disadvantages	References
Dummy nodes	Copy CSs as dummy nodes for multiple visits	Consistent with traditional framework of VRP model	Difficult to determine number of dummy nodes; increased problem complexity	Erdogan and Miller-Hooks (2012), etc. [14]
Arc detour	Change routing variable x_{ij} to x_{ijc}, where c represents a CS	No dummy nodes are used	Using a higher denominational variable; increasing problem-solving complexity	Koç and Karaoglan (2016) [23]; Leggieri and Haouari (2017) [23].
Simplified arc detour	Keep x_{ij}, while introducing another variable y_{ic}	Consistent use of two-dimensional variable	Introduced variable y_{ic} can be large if the CS number is large	Zuo et al. (2019) [3]
Fixed arc detour	Each arc (i,j) has a previously determined CS visit detour	CS selection is no longer needed; higher solving efficiency	Variable y_{ij} is introduced	This study

consider charging time as a definite value. Some researchers set up the same charging time for each CS visit (Conrad and Figliozzi, 2011; Erdogan and Miller-Hooks, 2012; Adler and Mirchandani, 2014; Montoya et al., 2015; Hof et al., 2017) [14,18,28–30], while others determined the charging time directly according to the battery level of the vehicle upon arrival at the station (Schneider et al., 2014; Goeke and Schneider, 2015; Schneider et al., 2015; Desaulniers et al., 2016; Hiermann et al., 2016; Lin et al., 2016; Szeto and Cheng, 2016) [5,15,21,31–34].

The partial charging policy adds charge time to the model as a decision variable. Most published works considered the time-varying LCF in their EVRP models with partial charging policy (Felipe et al., 2014; Sassi et al., 2015; Bruglieri et al., 2015; Schiffer and Walther, 2017; Desaulniers et al., 2016; Keskin and Catay, 2016) [5,16,35,38,39,41]. These two methods of full and partial charging policies with LCF only factor in the linear part of the charge function; however, the actual charge function is nonlinear (see Fig. 1). To describe the actual concave nonlinear charging function (NCF) of the partial charging policy during the charging process, Montoya et al. (2017) first used a piecewise linear approximation with four additional variables, and estimated the total cost of NCF to be 2.70 % cheaper than that of the LCF on average [17]. Subsequently, a new linearization method for the NCF was established by Zuo et al. (2019) with fewer additional variables than existing formulation [3].

The technology of battery swapping with a shorter fixed battery-swapping time has been applied in the charging time optimization [40]. Worley and Klabjan (2011) developed a dynamic programming model to determine the charging decision [42]. Mak et al. (2013) studied battery management at swapping stations [43]. Avci et al. (2014) analyzed the effectiveness of relieving carbon emissions and oil dependence by battery swapping [44]. Yang and Sun (2015) explored battery swapping stations, but their focus was more on how the facility location problem and did not take into account the time windows [20]. Chen (2016) incorporated battery swapping and time windows in their EVRP models [46].

2.3. Electricity/energy consumption model

An engine must provide sufficient power to propel an EV by overcoming the effects of aerodynamic, tire, drivetrain, ancillary, and kinetic/potential losses, and other factors (Marmaras et al., 2017) [47]. EVs traveling at low velocities have a higher (even double) travel range than those traveling at high velocities. The speed-varying travel range (SVR) is influenced by many factors because speed affects tire, aerodynamic, and kinetic/potential losses. Besides speed, other factors such as load and road gradient can directly affect the demand of drivetrains and indirectly affect the tire and kinetic/potential losses (Fiori et al.,

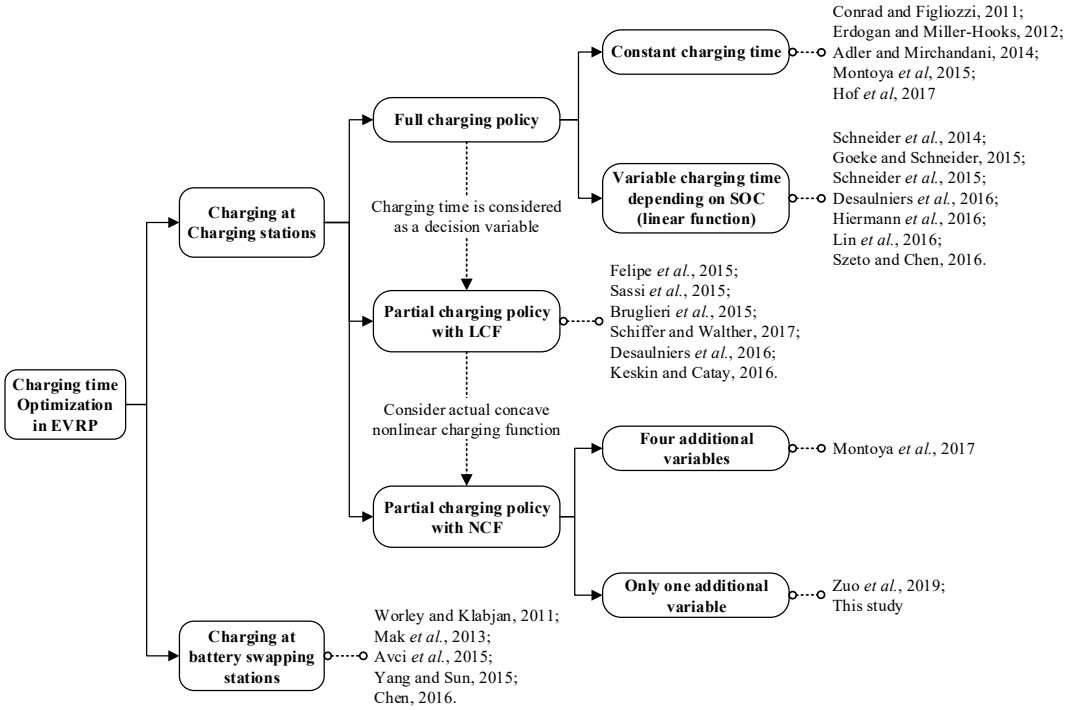


Fig. 2. Framework of charging time optimization in EVRP.

2016; Liu et al., 2017; Oriol et al., 2015) [37,49,50]. Moreover, local climate is also closely related with energy consumption. For instance, Javani et al.(2012) investigated that turning on the air conditioning can increase the energy consumption by up to a fifth [48]. In conclusion, the ECR per unit of traveled distance changes with the environmental condition during the journey, which in turn affects CS visit and charging time optimization. Accordingly, it is of practical significance to establish a dynamic ECR in the EVRP model.

With the popularization of EVs, research on the EV energy consumption model has witnessed rapid progress. Yuan et al. (2017) [51] introduced a physical-based statistical method to evaluate electricity consumption, considering the losses of battery, electric motor, driveline, brake, and load. Fetene et al. (2017) [53] used Big Data to evaluate electricity consumption, and demonstrated the nonlinear influence of driving patterns and weather conditions. Fiori et al. (2016) [50] developed an instantaneous energy consumption model for EVs for various transportation simulation software, such as smartphone eco-driving. However, these studies focused more on the evaluation of factors influencing the energy consumption of EVs, rather than developing analytical computation formulas. Consequently, they may be infeasible when applied to the EVRP model owing to ambiguous limitations and conditions.

This paper focuses on the electricity consumption models applied in the EVRP. Most EVRP researchers assume energy consumption as a linear function of travel distance, and thus conclude that the ECR is constant over each arc (Erdogan and Miller-Hooks, 2012; Felipe et al., 2014; Schneider et al., 2014) [14,16,31]. By considering the effects of practical factors such as speed, load, and road gradient, they assumed each arc as having its own ECR (Goeke and Schneider, 2015; Zuo et al., 2019) [3,15]. Recently, with consideration of the dynamic influence of speed and payload changes on the maximum remaining range of EVs, Xiao et al. (2019) [7] used the empirical regression method to devise an improved ECR model with continuous decision variables of speed and load. In this study, a dynamic ECR model was developed by summing up all practical consumption that lead to aerodynamic, tire, drivetrain, ancillary, and kinetic/potential losses. Fig. 3 summarizes the ECR models developed to solve the EVRP and the corresponding literature.

2.4. EVRP with practical factors

Researchers must consider the unique characteristics and various technical constraints of the EVRP, including vehicle capacity/load capacity, vehicle battery capacity, vehicle energy consumption, available CSs, charging technology used, CS opening time window, CS capacity limitation, grid capacity limitation, instant electricity price and charging cost, etc. Among them, EVRP-C (EVRP with charging), EVRP-TW problem (EVRP with time window), and EVRP-MF problem (EVRP with mixed fleet) are the main research directions, the framework of which are summarized along with the corresponding studies in Fig. 4.

2.4.1. EVRP with charging (EVRP-C)

On the one hand, the EVRP-C study can be divided into three categories by charging location: charging at customer location, charging at depots, and charging at CSs. Conrad and Figliozzi (2011) [18] assumed that EVs are charged at the same time when customers complete their services. In the early stages of EVs entering the logistics services, there were few public CSs available; therefore, many vehicles could only be charged at depots. Therefore, charging at depots has drawn the attention of researchers. By assuming vehicles can charge at depots, Goeke and Schneider (2015) [15] and Penna et al. (2016) [25] introduced the multi-depot EVRP with inter depot routes (MD-EVRP-I) problem. Most EVRP-C researchers considered charging at a CS midway. However, insufficient number of CSs may prompt EV drivers to take a detour. Hence, how to visit the nearest CS with the shortest detour (or the lowest cost) are important to solving the EVRP-C problem (Erdogan and Miller-Hooks, 2012; Yang and Sun, 2015; Barco et al., 2017) [14,20,52].

On the other hand, limited charging capacity at each CS is also a great challenge to EVRP-C research. With consideration of capacity limitation, Ding et al. (2015) [54] proposed a heuristic method with simple charging time adjustment processes, and Froger et al. (2017) [55] developed the route-first assemble-second algorithm. Kchaou-Boujelben and Gicquel (2020) [56] and Soysal et al. (2020) [57] studied the charging station location problem under stochastic energy status.

2.4.2. EVRP with time window (EVRP-TW)

Unlike shorter customer service time, longer charging time cannot be ignored. Moreover, CSs, depots, etc. may often have time window restrictions. Hence, the EVRP-TW research must ensure that all customers are served efficiently at the lowest cost, by managing charging time and delivery time, to obtain the optimal path planning that meets the time window constraints, service requirements, and battery constraints (Schneider et al., 2014; Booth and Beck, 2019) [31,59].

In particular, each customer has their earliest and latest service start time, and an EV can only visit the customer within this time window. If the EV arrives earlier than the earliest service, it is required to wait at the customer's; if it arrives after the latest service start time, this solution would be infeasible (hard time window) (Schneider et al., 2014; Hiermann et al., 2016; Xiao et al., 2019; Zuo et al., 2019) [3,7,21,31] or feasible with a corresponding penalty (soft time window) (Iqbal and Rahman, 2013) [22].

2.4.3. EVRP with mixed fleet (EVRP-MF)

Logistics companies typically have different types of conventional vehicles and EVs, with varying travel range, load, and cost. For instance, large EVs may be able to complete delivery without charging midway, but the fixed cost is higher. Small EVs can reduce the fixed cost but have longer charging time and more detours owing to their short travel range. Accordingly, a mixed fleet of different vehicle types can optimize the total costs for logistics companies (Hiermann et al., 2016; Rezgui et al., 2019; Golden et al., 1984), [21,58,61].

Existing mixed fleets in the literature can be classified into three modes—mixed fleet of different types of EVs, mixed fleet of the same type of EVs and oil-fueled vehicles, and mixed fleet of different types of

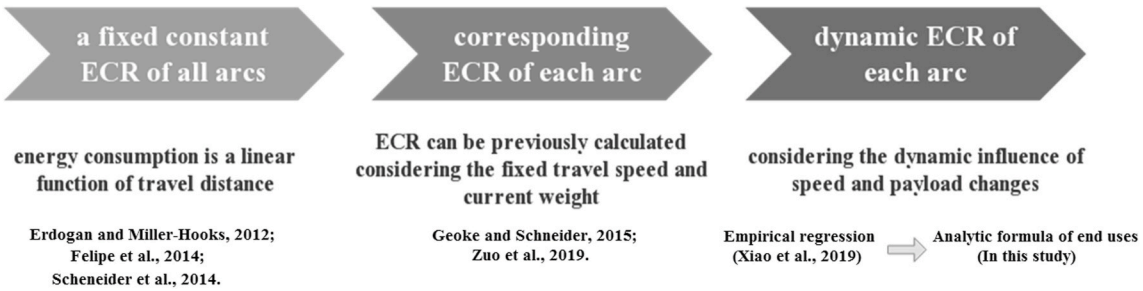


Fig. 3. Development of ECR models in EVRP.

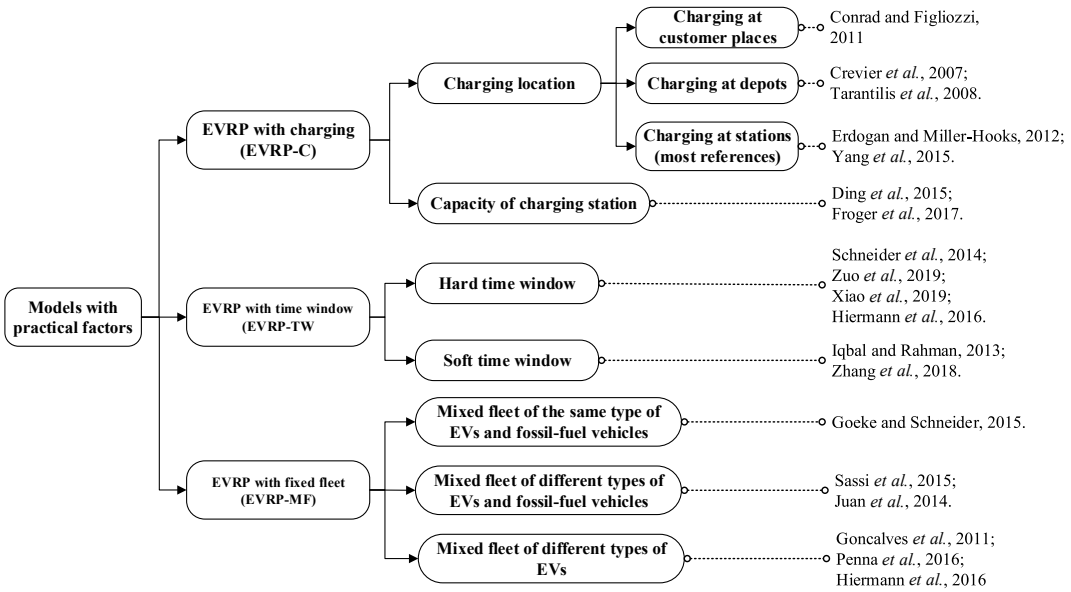


Fig. 4. Framework of models with practical factors in EVRP.

EVs and oil-fueled vehicles. The EVRP-MF problem was first proposed by Gonçalves et al. (2011) [62] on the premise of arbitrary charging location. Based on this assumption, Van et al. (2013) [65] introduced the EVRP-MF problem with time windows. Later, Hiermann et al. (2016) [21] first extended the assumption of charging location to accessible public CSs and Zuo et al. (2017) [64] considered both the customer demand and the vehicle capacity. In the above four models, the fleet was composed of different types of EVs. Goeke and Schneider (2015) [15] studied the EVRP of a hybrid fleet of the same type of EVs and fossil fuel vehicles, and calculated the corresponding energy consumption cost for each type of vehicle. Juan et al. (2014) [60] studied the EVRP-MF by considering different types of hybrid and electric vehicles as multiple driving range. Then Sassi et al. (2015) [34] further expanded the EVRP-MF model to study the mixed fleet of different types of EVs (different battery capacities and operating costs) and fossil fuel vehicles.

2.5. Solution approaches

This sub-section provides review of the existing literature on the EVRP solution. The main solution approaches can be divided into the exact method, problem-specified heuristic method, metaheuristic method, and hybrid heuristic method. Fig. 5 illustrates the four types of solution approaches along with the corresponding literature.

2.5.1. Exact algorithm

Exact algorithm always finds the optimal solution to an optimization problem by using mathematical model or data structure. Desaulniers et al. (2016) [5] used an exact algorithm of branch-price-and-cut to seek

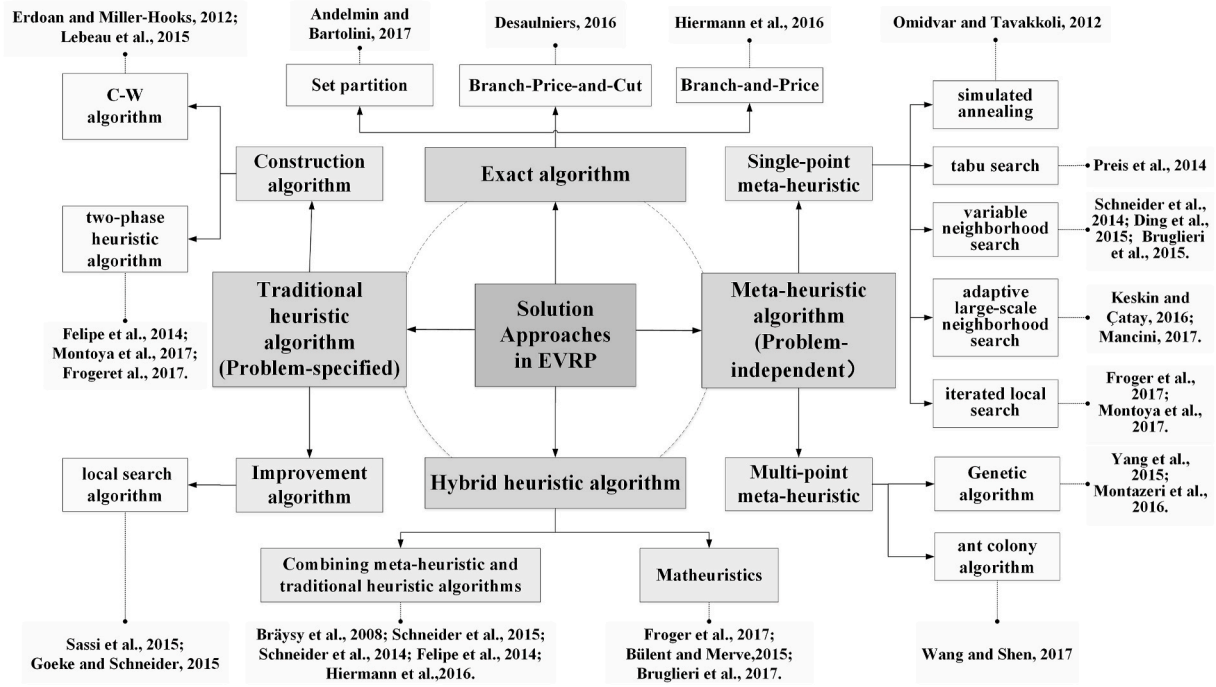


Fig. 5. Framework of solution approaches and literature in EVRP.

the optimal solution for each of four EVRP-TW variants. Hiermann et al. (2016) [21] through branch-and-price optimized logistics routes by considering charging time and locations. The set partition exact algorithm was proposed by Andelmin and Bartolini (2017) [66] to solve the green vehicle routing problem, adding valid inequalities including k-path cuts.

The optimal solution is usually obtained by an exact algorithm in an acceptable time in small-scale instances. However, the time required for the exact algorithm to obtain the optimal solution increases exponentially as the problem sizes increase linearly. Various types of heuristic algorithms are adopted by EVRP research works to obtain a feasible solution for large-scale instances within an acceptable timeframe.

2.5.2. Traditional heuristic algorithm (problem-specified)

Traditional heuristic algorithms are problem-specific programs that improve solutions according to the special structure or nature of the problem. In general, the traditional heuristic method is the most basic solution method in the EVRP study. Its main principle is to start from the initial solution and find a better one in its neighborhood as the current solution, then repeat the process until no better solution can be updated. Thus, the algorithm can obtain an effective (but not necessarily optimal) solution within an acceptable timeframe. Heuristic algorithms can be classified into construction algorithms and local improvement algorithms.

The construction algorithm starts with "zero," runs a series of steps, and adds part of the solution at a time. It repeats this process until it has obtained a complete, feasible solution. The Clarke and Wright savings algorithm (C–W algorithm) is a type of typical construction algorithm. Erdogan and Miller-Hooks (2012) [14] used an improved C–W algorithm to solve the routing problem of renewable fuel vehicles, and a feasible route was created by inserting CSs, merging paths, and removing redundant CSs. Lebeau et al. (2015) [67] proposed the EVRP–FMTW model by using the improved C–W algorithm. Another widely used construction algorithm is the two-phase heuristic algorithm, which has been successfully applied in various EVRP models. EVRP-FM with time-dependent charging cost proposed by Felipe et al. (2014) [68] was solved by the two-stage heuristic algorithm. The algorithm was also applied to EVRP models with nonlinear charging (Montoya et al., 2017) [17] and limited charging capacity (Froger et al., 2017) [55].

Unlike the construction algorithm, the local improvement algorithm starts from a feasible solution and iteratively performs the improvement operation until a certain termination condition is satisfied. The local improvement algorithm includes various types of specific algorithms, the most classic one being the local search algorithm, which has been already widely combined with other algorithms in various EVRP studies (Sassi et al., 2015; Goeke and Schneider, 2015) [15,34].

2.5.3. Metaheuristic algorithm (problem-independent)

Metaheuristic algorithm is a type of problem-independent heuristic that does not rely on the specific conditions of a problem and can be applied to a wider range of aspects. The basic principle of a metaheuristic algorithm is to start from the initial solution and achieve a better solution through repeated local perturbations of the current solution. It can be divided into single-point and multi-point metaheuristic algorithms.

The single-point metaheuristic algorithm is based on a single solution in the optimization process. The commonly used single-point metaheuristic algorithms in the EVRP include simulated annealing algorithm, tabu search algorithm, variable neighborhood search algorithm, adaptive large-scale neighborhood search algorithm, and iterated local search algorithm. In the process of solving, the simulated annealing algorithm jumps out of the local optimal with a certain probability, and finally tends to become the global optimal (Omidvar and Tavakkoli, 2012) [69]. The tabu search algorithm introduces tabu tables and tabu rules to avoid falling into local optimization, to achieve a better solution (Preis et al., 2013) [70]. The variable neighborhood search algorithm extends the scope of search by systematically changing the neighborhood during the search process (Schneider et al., 2014; Ding et al., 2015; Bruglieri et al., 2015) [31,38,54]. The adaptive large-scale neighborhood search algorithm can obtain a better solution by destroying and reconstructing partial solutions in the iterative process (Keskin and Çatay, 2016) [41]. The iterative local search algorithm is realized by adding perturbations to the local optimal solution and repeating the local search (Froger et al., 2017; Montoya et al., 2017) [17,55].

The multi-point metaheuristic algorithm is based on multiple solution vectors in the solution. Genetic algorithm (GA) and ant colony algorithm are typical multi-point metaheuristic algorithms. GA was used by Yang et al. (2015) [45] to solve the EVRP model with receipt and delivery. The strategy of parallel charging of vehicles was optimized based on the application of GA (Montazeri et al., 2006) [63]. Wang and Shen (2017) [73] used the ant colony algorithm to solve the scheduling problem of electric buses.

2.5.4. Hybrid heuristic algorithm

In many EVRP studies, the hybrid heuristic algorithm formed by combining various metaheuristic and heuristic algorithms. Structurally, the hybrid method can be a simple parallel (the two methods are used in sequence) or an inseparable combination. For instance, some simple heuristic algorithms are generally used to obtain the initial solution during the execution of the metaheuristic algorithm. The C–W saving algorithm was used to obtain the initial solutions of the EVRP-MFTW model proposed by (Bräysy et al., 2008) [72] and the EVRP-C model proposed by (Schneider et al., 2015) [32]. Moreover, Schneider et al. (2014) [31] proposed the EVRP–TW model by combining the variable neighborhood search and tabu search algorithms, adopting a dynamic punishment mechanism to guide the search feasible solution. Besides the C–W saving algorithm, Schneider et al. (2015) [32] also combined the adaptive large-scale neighborhood search and variable neighborhood search algorithms to solve vehicle routing problems with intermediate stops. Felipe et al. (2014) [16] combined the simulated annealing and iterated local search algorithms to solve the EVRP model with partial recharges. The most representative study proposed by Hiermann et al. (2016) [21] used a branch-and-price algorithm for small-scale instances and combined the adaptive large-scale neighborhood search and iterated local search algorithms for large-scale instances.

In recent years, matheuristics, the combination of mathematical programming and the matheuristics, has drawn research attention (Boschetti et al., 2009; Doerner and Schmid, 2010; Archetti and Speranza, 2014) [74–76]. Matheuristics combines the advantages of the strong robustness of mathematical programming and the high efficiency of a metaheuristic algorithm. It has been applied in various EVRP-based research works. A matheuristic of sorting before clustering was proposed to solve the EVRP-C model considering a nonlinear recharging function (Froger et al., 2017) [55]. In the first stage, an iterative local search algorithm was adopted to optimize the initial solution, ignoring the CS capacity limitation. In the second phrase, the branch-and-bound algorithm was used to shear infeasible solutions and obtain the final solution. A matheuristic of combining the adaptive large neighborhood search algorithm and the exact method was developed by Bülent and Merve (2016) [36] to solve the EVRP-TW model. Bruglieri et al. (2017) [71] proposed a three-phase matheuristic to route a fleet of EVs. The first two phases were based on MILP to generate feasible solutions.

2.6. Review summary

In Table 2, a summary of various critical factors recognized so far for the EVRP modeling are provided, as well as related typical references including this study. The driving range of an EV is determined and influenced by many factors, including features of specific EV (see section 3.1), speed of travel flow on road, load of vehicle, status of air condition, road condition such as the average numbers of L or U turns and traffic lights per mile, and outside temperature. The charge modeling for an EV

Table 2
Summary of critical factors in EVRP modeling.

CS visit	Variable factors of dynamic driving range							Fixed range	Charge modeling				Battery swapping	Reference examples
	EV's Features	Travel speed	Load	A/C	L/U turns	Traffic lights	Outside temp.		No	Linear	Nonlinear			
											Empirical	Theoretical		
✓								✓	✓					Erdogan and Miller-Hooks (2012), Koç and Karaoglan (2016), etc.
✓								✓					✓	Mak et al. (2013); Chen (2016)
✓								✓		✓				Schiffer and Walther, 2017; Desaulniers et al., 2016; etc.
✓								✓			✓			Montoya et al., 2017; Pelletier et al., 2017
✓		✓	✓						✓					*Xiao* et al. *(2019)*
✓								✓			✓			*Zuo* et al. *(2019)*
✓	✓	✓	✓	✓							✓			This study

involves three strategies: linear, nonlinear, and battery swapping.

As a conclusion of the literature review part, three aspects, as seen in Table 2 for the blank columns *L/U turns*, *Traffic lights*, *Outside temperature*, and *Theoretical charging function*, still remains challenges in the EVRP modeling, which may be critical in some applications, as follows.

(1) To consider the effects of road conditions on the dynamic driving range, such as the average number of L or U turns per mile and the average number of traffic lights per mile, which may cause extra kinetic/potential energy loss due to velocity changing.
(2) To consider the effect of outside temperature on the dynamic driving range. The battery thermal management system (BTMS) consumes more electric energy to heat or cool the battery when the environmental temperature is too cold or high during operation.
(3) To develop a theoretical model, rather than empirical models based on statistical data, for the nonlinear electricity charging function, considering charging sensitive factors such as SOC, charging voltage, temperature, time, etc.

3. Energy consumption rate (ECR) modeling for an electric vehicle

3.1. ECR model of electric vehicle

The ECR of an EV indicates the amount of battery electricity consumed for per unit of traveled distance. Tesla Motor CTO JB Straubel broke down the four different end uses, namely, Aerodynamic Loss, Tire Loss, and Drivetrain Loss, and Ancillary Loss, of battery capacity consumed at different constant driving velocities (https://www.tesla.com/blog/roadster-efficiency-and-range, http://www.solarjourneyusa.com/EVdistanceAnalysis5.php). With this backbone, a dynamic ECR with varying speed and load as the decision variables is applied to the EVRP model of this study, where the battery capacity includes the following five end uses.

(1) Aerodynamic loss

Aerodynamic loss is an important part of the ECR when the EV is traveling at high velocities. The force of air resistance acting against the relative motion of an object is calculated as $\rho AC_d v^2/2$, where ρ is the density of air (kg/m^3), A is the frontal area (in m^2), C_d is the coefficient of drag, and v is the velocity (in m/s). Thus, when an EV is traversing a distance d (in m), its ECR (in W-h/km) due to aerodynamic losses is calculated by

$$ECR_{air} = \frac{1}{2}\rho AC_d v^2 d \cdot 1000 \Big/ 3600 \tag{1}$$

where 1000/3600 is used to convert the unit of ECR from J/m to W-h/km.

(2) *Tire loss*

The tire loss is the energy consumed for overcoming the rolling resistance of a road surface on the tires of a vehicle. The force of rolling resistance is determined by $N{\cdot}C_r$, where N is the vertical force of wheels on the road surface and C_r is the coefficient of rolling resistance determined by tire parameters and road conditions. The ECR (in Watt-hour/kilometer) due to tire loss when a vehicle is traveling a distance d (in m) is calculated by

$$ECR_{tire} = (W+f)gC_r \cos(\Phi) d \cdot 1000/3600 \tag{2}$$

where W is the curb-weight of the vehicle, f is the payload (in kg), g is the acceleration due to gravity, C_r is the rolling resistance coefficient, and Φ is the road angle.

(3) Drivetrain loss

Drivetrain loss is any loss resulting from the process of converting energy in the battery to torque at the wheels of the vehicle. Here this loss is considered as a polynomial regression of data published by Straubel:

$$P_{Dr} = \alpha_{Dr} v^3 + \beta_{Dr} v^2 + \gamma_{Dr} v + C_{Dr} \tag{3}$$

where P_{Dr} is the power of drivetrain loss (in W), the constant CDr represents the power usage of the complete drivetrain system when the vehicle is not moving, coefficients α_{Dr}, β_{Dr}, and γ_{Dr} are constants related to each vehicle, and v is the velocity (in m/s). The ECR (in W-h/km) due to drivetrain loss for traveling a distance d (in m), is calculated by

$$\begin{aligned} ECR_{Dr} &= P_{Dr} d/v \\ &= \left(\alpha_{Dr} v^3 + \beta_{Dr} v^2 + \gamma_{Dr} v + C_{Dr}\right) d/v \cdot 1000/3600 \end{aligned} \tag{4}$$

(4) Ancillary loss

Ancillary losses are considered 'all other' electrical loads in the vehicle, including the air conditioning (A/C) system, onboard computer, audio, and lights, as well as systems necessary to regulate battery temperature. The total power of ancillary losses is expressed as $P_{Anc} = P_{A/C} + P_{others}$, where $P_{A/C}$ is the power (in W) of the A/C system and P_{others} is the summarization of powers of all other ancillary electric systems in the vehicle (in W). The ECR (in W-h/km) due to ancillary losses in an EV

covering a distance d (in m), which takes time d/v seconds, can be calculated by

$$ECR_{ancillary} = \left(P_{A/C} + P_{others}\right) d/v \cdot 1000/3600 \quad (5)$$

weight and payload of vehicle, respectively.

By summing up the aerodynamic, tire, drivetrain, ancillary, and kinetic/potential losses, the ECR of an EV is formulated as follows.

$$\begin{aligned} ECR &= ECR_{air} + ECR_{tire} + ECR_{drv} + ECR_{anc} + ECR_{kp} \\ &= (W+f)[gC_r \cos(\Phi) + g \sin(\Phi) + a] d \cdot 1000/3600 \\ &+ \left(\frac{1}{2}\rho A C_d v^2 + \alpha_{Dr} v^2 + \beta_{Dr} v + \gamma_{Dr} + C_{Dr} v^{-1} + P_{A/C} v^{-1} + P_{others} v^{-1}\right) d \cdot 1000 \Big/ 3600 \end{aligned} \quad (7)$$

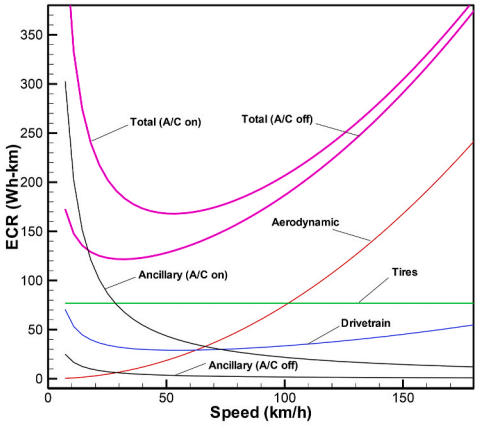
(5) Kinetic/potential loss

Kinetic/potential loss is the energy lost from the battery while converting chemical energy into kinetic/potential energy owing to varying velocities and terrain heights. The instantaneous power for kinetic/potential losses (in W-h/km) is calculated by

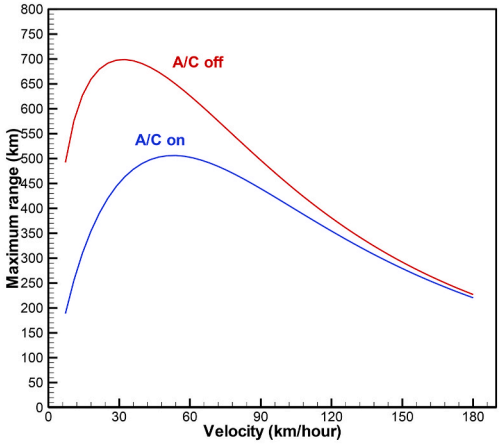
$$ECR_{kp} = (W+f)[g \sin(\Phi) d + a] \cdot 1000/3600 \quad (6)$$

where d is the traveled distance, a is the vehicle acceleration, g is the acceleration due to gravity, Φ is the road angle, and W and f are the curb-

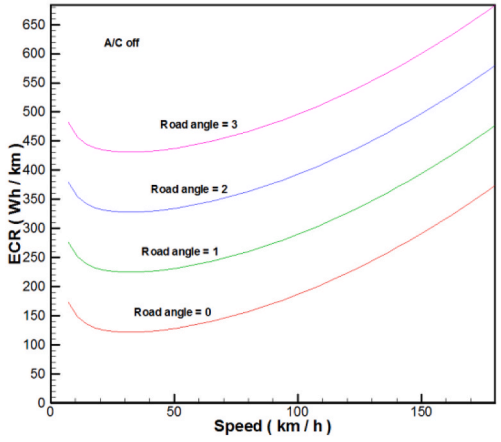
Thus, the maximum range (noted as L in km) of an EV with battery capacity C (in kWh) can be calculated by $L = 1000 \cdot C/ECR$. When a no-load vehicle is traveling at a constant speed (zero acceleration) on a road with zero angle, the ECR and maximum range are non-linearly affected by the travel speed, as shown in Fig. 6(A) and Fig. 6(B), respectively. The road angle can also affect the ECR and maximum range considerably, as shown in Fig. 6(C) and (D), when an EV is driving on a road with an inclination angle. The parameter values used to draw the ECR curves are given in Table 3 based on an EV model, part of which were collected from JB Straubel's post and from the manual book of Tesla Model S 85.



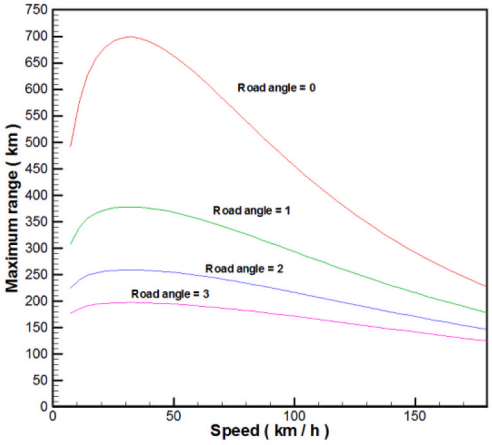
(A) Different usages of energy losses (Φ=0, α=0)



(B) Range vs. speed(Φ=0, α=0)



(C) ERC affected by road angles (α=0)



(D) Range vs. speed (α=0)

Fig. 6. Energy losses and range vs. speed.

Table 3
Parameter values of the ECR function of an EV model.

Notation	Description	Value
ρ	Air density (kg/m^3)	1.293
A	Frontal surface area (m^2)	2.341
C_d	Coefficient of aerodynamic drag	0.23
W	Curb-weight (kg)	2169
g	Acceleration due to gravity (m/s^2)	9.81
C_r	Coefficient of rolling resistance	0.013
α_{Dr}	Coefficients of drivetrain related to speed^3	0.0448
β_{Dr}	Coefficients of drivetrain related to speed^2	0.2502
γ_{Dr}	Coefficients of drivetrain related to speed	65.542
C_{Dr}	Power usage of drivetrain system when vehicle is not moving (W-h)	375
$P_{A/C}$	Power of A/C system in vehicle (W)	2000
P_{other}	Power of all other ancillary electric systems in vehicle (W)	180
Φ	Road angle	0, 1, 2, 3
a	Vehicle acceleration	0
d	Traveled distance (m)	1000
C	Battery capacity of vehicle (kWh)	85

3.2. Linearization of ECR model

To guarantee the to-be proposed EVRPTW model (in Section 5) solvable to Branch & Bound algorithms embedded in many existing MIP solvers like CPLEX, in this section, the nonlinear components of the ECR model established above are linearized by proposing the linearization techniques of the nonlinear quadratic function and inverse proportional function, as follows.

The ECR function in Eq. (7) can be further simplified as Eq. (8).

$$e(v,f) = \alpha v^2 + \beta v + \gamma + \phi v^{-1} + \chi f \tag{8}$$

where $e(v, f)$ represents the ECR function, v is the velocity variable (km/h), and f is the payload variable (kg). The coefficients in Eq. (8) are expressed as

$$\alpha = (0.5\rho A C_d + \alpha_{Dr})/3.6$$

$$\beta = \beta_{Dr}/3.6$$

$$\gamma = (W(gC_r\cos(\Phi) + g\sin(\Phi) + a) + \gamma_{Dr})/3.6$$

$$\phi = \left(C_{Dr} + P_{A/C} + P_{others}\right)/3.6\ , \text{ and}$$

$$\chi = (gC_r\cos(\Phi) + g\sin(\Phi) + a)/3.6$$

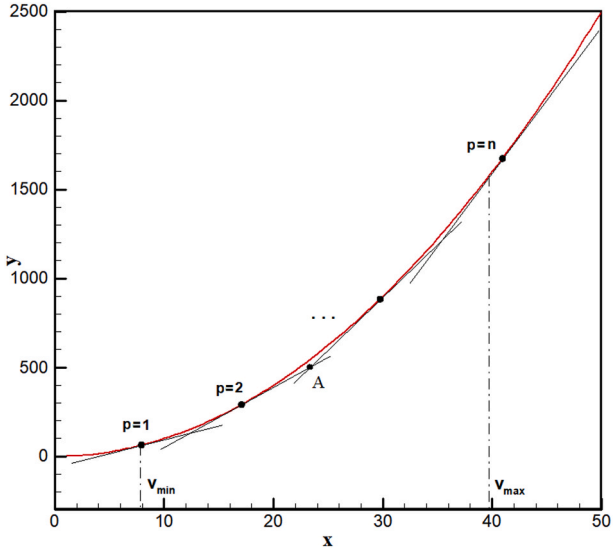


Fig. 7. Linearization of nonlinear curve y = x^2 using tangential lines.

With the values of parameters in Table 3, $\alpha = 0.109130132$, $\beta = 0.069498389$, $\gamma = 95.04300093$, $\phi = 709.7222222$ (for A/C on), $\phi = 154.1666667$ (for A/C off), and $\chi = 0.035425$. Let $e_1(v) = \alpha v^2$, $e_2(v) = \beta v$, and $e_4(v) = \phi v^{-1}$, such that Eq. (8) can be written separately as

$$e(v,f) = e'(v) + \chi f = e_1(v) + e_2(v) + \gamma + e_4(v) + \chi f \tag{9}$$

where $e'(v)$ represents the ECR of a vehicle with no load and χf represents the additional ECR due to the vehicle's payload. It can be observed that there are two nonlinear terms in Eq. (9), i.e., the first term $e_1(v) = \alpha v^2$ and the fourth term $e_4(v) = \phi v^{-1}$.

3.2.1. Linearization of quadratic component

In this sub-section, the piecewise linearization technique of quadratic component is first proposed and applied to the linearization of ECR model. The basic nonlinear curve of $e_1(v) = \alpha v^2$, i.e., $y = x^2$, as shown in Fig. 7, can be surrogated linearly by a set Q of tangential lines. A tangential line p, $p \in Q$, is expressed as $y = k_p x + b_p$, where k_p and b_p represent the slop and intercept, respectively. Then, the following inequality is used to bound y and x as follows.

$$y \geq k_p x + b_p \quad \forall p \in Q, x \in [x_{\min}, x_{\max}] \tag{10}$$

The maximum deviation of bounding y by Eq. (10) lies on the cross-point A of two neighboring tangential lines (Fig. 7). For any two neighboring tangential lines, say pth and $(p+1)^{\text{th}}$ lines, their intersection, denoted by (x', y'), is calculated by the points $x' = (x_p + x_{p+1})/2$ and $y' = x_p x_{p+1}$. The recurrence formula in Eq. (11) can be used to calculate the tangential points of lines that guarantee the maximum approximation error will be less than a given maximum range, ε%, i.e., $\frac{x'^2 - y'}{x'^2} \times 100 \leq \varepsilon$, as follows.

$$\begin{cases} x_p = \mu^{p-1} v_{\min} \\ y_p = x_p^2 \end{cases} \quad \forall p = 1, 2, \ldots, \eta \tag{11}$$

where $v_{\min}$ is the minimum speed, η is the number of tangential lines, and $\mu = (1 + \sqrt{\varepsilon})/(1 - \sqrt{\varepsilon})$. To guarantee the maximum speed $v_{\max}$ is covered by the tangential lines, i.e., $v_{\max} \leq v_\eta$, the minimum integer η can be calculated by Eq. (12).

$$\eta = \left\lceil \frac{\ln v_{\max} - \ln v_{\min}}{\ln \mu} + 1 \right\rceil \tag{12}$$

where $\lceil a \rceil$ indicates the integer that is right above a.

The slope and intercept of the pth tangential line on the curve $y = x^2$ can be deduced as follows.

$$\begin{cases} k_p = 2x_p = 2\mu^{p-1} x_{\min} \\ b_p = -x_p^2 = -\mu^{2(p-1)} x_{\min}^2 \end{cases} \quad \forall p = 1, 2, \ldots, \eta \tag{13}$$

Thus, the following inequalities in Eq. (14) can be used to linearize $e_1(v) = \alpha v^2$, as follows.

$$e_1 \geq \alpha k_p v + \alpha b_p, \quad \forall p \in 1, 2, \ldots, \eta \tag{14}$$

In Table 4, the values of k_p and b_p with respect to the different ε values are provided for the speed range [20,120] km/h. Readers may directly refer to these data to linearize their own models.

3.2.2. Linearization of inverse proportional component

The third term $e_4(v) = \phi v^{-1}$ has the same mathematical expression as $t_{i\,j} = D_{ij}/v_{ij}$. The linearization of bounding travel time and travel speed was first proposed by Xiao et al. in 2019 and the technique is introduced as follows:

For a given arc (i, j) with a known distance D_{ij}, the travel time t_{ij} and travel speed v_{ij} have a nonlinear relationship, i.e., $t_{ij} = D_{ij}/v_{ij}$, and it can be approximated by using a set of tangential lines, denoted by $P = \{p_1,$

Table 4
Data of kp and bp for speed range [20, 120] km/h.

ε	p	k_p	b_p	ε	p	k_p	b_p
5.0 %	1	2.000000E+00	−1.000000E+00	0.1 %	1	2.000000E+00	−1.000000E+00
	2	6.304057E+01	−9.935284E+02		2	4.261243E+01	−4.539549E+02
	3	9.935284E+01	−2.467747E+03		3	4.539549E+01	−5.151876E+02
	4	1.565815E+02	−6.129442E+03		4	4.836031E+01	−5.846799E+02
	5	2.467747E+02	−1.522444E+04		5	5.151876E+01	−6.635457E+02
2.0 %	1	2.000000E+00	−1.000000E+00		6	5.488350E+01	−7.530495E+02
	2	5.317725E+01	−7.069551E+02		7	5.846799E+01	−8.546263E+02
	3	7.069551E+01	−1.249464E+03		8	6.228658E+01	−9.699045E+02
	4	9.398482E+01	−2.208287E+03		9	6.635457E+01	−1.100732E+03
	5	1.249464E+02	−3.902899E+03		10	7.068824E+01	−1.249207E+03
	6	1.661076E+02	−6.897935E+03		11	7.530495E+01	−1.417709E+03
	7	2.208287E+02	−1.219133E+04		12	8.022319E+01	−1.608940E+03
	8	2.935766E+02	−2.154680E+04		13	8.546263E+01	−1.825965E+03
1.0 %	1	2.000000E+00	−1.000000E+00		14	9.104427E+01	−2.072265E+03
	2	4.888889E+01	−5.975309E+02		15	9.699045E+01	−2.351787E+03
	3	5.975309E+01	−8.926078E+02		16	1.033250E+02	−2.669013E+03
	4	7.303155E+01	−1.333402E+03		17	1.100732E+02	−3.029029E+03
	5	8.926078E+01	−1.991872E+03		18	1.172622E+02	−3.437606E+03
	6	1.090965E+02	−2.975512E+03		19	1.249207E+02	−3.901295E+03
	7	1.333402E+02	−4.444901E+03		20	1.330794E+02	−4.427530E+03
	8	1.629713E+02	−6.639914E+03		21	1.417709E+02	−5.024747E+03
	9	1.991872E+02	−9.918884E+03		22	1.510301E+02	−5.702522E+03
	10	2.434510E+02	−1.481710E+04		23	1.608940E+02	−6.471719E+03
0.5 %	1	2.000000E+00	−1.000000E+00		24	1.714021E+02	−7.344671E+03
	2	4.608729E+01	−5.310096E+02		25	1.825965E+02	−8.335374E+03
	3	5.310096E+01	−7.049280E+02		26	1.945221E+02	−9.459709E+03
	4	6.118198E+01	−9.358088E+02		27	2.072265E+02	−1.073570E+04
	5	7.049280E+01	−1.242309E+03		28	2.207606E+02	−1.218381E+04
	6	8.122055E+01	−1.649194E+03		29	2.351787E+02	−1.382725E+04
	7	9.358088E+01	−2.189345E+03		30	2.505384E+02	−1.569237E+04
	8	1.078222E+02	−2.906408E+03				
	9	1.242309E+02	−3.858327E+03				
	10	1.431366E+02	−5.122021E+03				
	11	1.649194E+02	−6.799606E+03				
	12	1.900173E+02	−9.026640E+03				
	13	2.189345E+02	−1.198308E+04				
	14	2.522525E+02	−1.590783E+04				

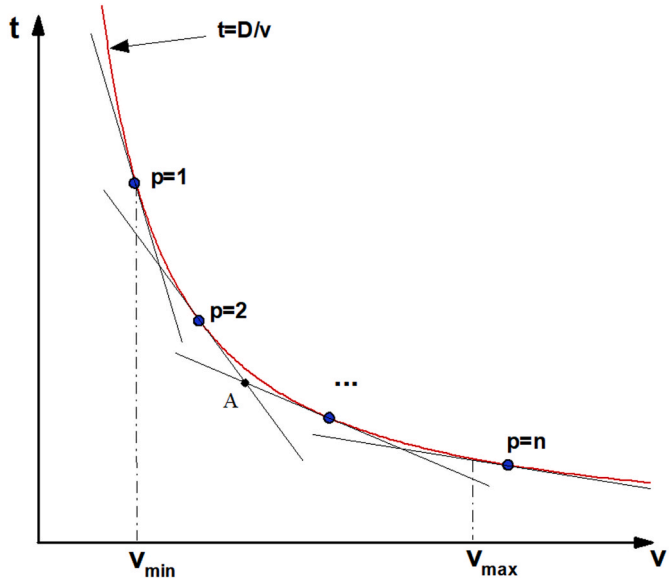


Fig. 8. Linearization of time–speed relation using tangential lines.

$p_2, \ldots\}$, starting from the minimum speed ($v_{\min}$) to the maximum speed ($v_{\max}$), as shown in Fig. 8. Each tangential line $p \in P$ is defined as $t_{ij} = K_p^{(i,j)} v_{ij} + B_p^{(i,j)}$ with slope $K_p^{(i,j)}$ and intercept $B_p^{(i,j)}$ passing through the tangential point (v_p, t_p) on the curve $t_{ij} = D_{ij}/v_{ij}$. If the objective function involves minimizing the cost positively with respect to the travel time, then the travel time t_{ij} can be restricted to a lower bound value by a set of linear constraints in Eq. (15) with respect to the travel speed v_{ij}, as follows.

$$t_{ij} \geq K_p^{(i,j)} v_{ij} + B_p^{(i,j)}, \quad \forall p \in P \tag{15}$$

The maximum deviation of bounding the travel time t_{ij} using Eq. (15) lies on the intersection of two neighboring tangential lines, as the point A shown in Fig. 8. Thus, for any two neighboring tangential lines, say the pth and $(p+1)^{\text{th}}$ lines, it can be deduced that the coordinates of the intersection are $(\frac{2v_p v_{p+1}}{v_p + v_{p+1}}, \frac{2D_{ij}}{v_p + v_{p+1}})$. Let t_{ij}' be the approximated value for travel time t_{ij}, then the approximation error rate (AER) is calculated by $\frac{|t_{ij}' - t_{ij}|}{t_{ij}} \times 100\%$. Apparently, the increase in the number of tangential lines in the approximation will lead to a lower AER. The recurrence formula in Eq. (16) can be used to calculate the tangential points required to ensure the maximum AER is less than a given AER ε%, i.e., $\frac{|t_{ij}' - t_{ij}|}{t_{ij}} \times 100\% \leq \varepsilon$, as follows.

Table 5
Required minimum number of secant lines for speed range [20, 120] km/h.

Maximum allowed deviation ε	5.0 %	2.00 %	1.00 %	0.50 %	0.10 %	0.05 %	0.01 %
Minimum number of secant lines n	5	7	10	14	30	42	91

$$\begin{cases} v_1 = v_{\min}, t_1 = D_{ij}/v_{\min} & \forall p = 1 \\ v_{p+1} = \mu v_p, t_{p+1} = D_{ij}/(\mu v_p) & \forall p = 2, 3, ..., \eta \end{cases} \tag{16}$$

where $v_{\min}$ is the minimum speed, η is the number of tangential lines, and $\mu = (1 + \sqrt{\varepsilon})/(1 - \sqrt{\varepsilon})$. The minimum integer η that guarantees the maximum speed $v_{\max}$ is covered by tangential lines, i.e., $v_{\max} \le v_\eta$, can be calculated by the following equation.

$$\eta = \left\lceil \frac{\lg(v_{\max}/v_{\min})}{\lg\mu} + 1 \right\rceil \tag{17}$$

where $\lceil a \rceil$ indicates the integer that is right above a. In Table 5, the values of η with respect to different values of ε are provided for the speed range [20 km/h, 120 km/h].

By using Eq. (16), the tangential point (v_p, t_p) of any line in P, e.g., the pth tangential line, can be deduced by the equation in Eq. (18).

$$\begin{cases} v_p = \mu^{p-1} v_{\min} \\ t_p = D_{ij}/(\mu^{p-1} v_{\min}) \end{cases} \quad \forall p = 1, 2, ..., \eta \tag{18}$$

Thus, the slope $K_p^{(i,j)}$ and intercept $B_p^{(i,j)}$ of the pth tangential line on the curve $t_{ij} = D_{ij}/v_{ij}$ for traveling the given arc (i, j) can be deduced as follows.

$$\begin{cases} K_p^{(i,j)} = -D_{ij}\mu^{2-2p}/v_{\min}^2 \\ B_p^{(i,j)} = 2D_{ij}\mu^{1-p}/v_{\min} \end{cases} \quad \forall p = 1, 2, ..., \eta \tag{19}$$

Let $K_p = -\mu^{2-2p}/v_{\min}^2$, $B_p = 2\mu^{1-p}/v_{\min}$, $K_p^{(i,j)} = K_p D_{ij}$, and $B_p^{(i,j)} = B_p D_{ij}$. Thus, Eq. (1) can be substituted by Eq. (20) as follows.

$$t_{ij} \ge D_{ij}K_p v_{ij} + D_{ij}B_p, \forall p \in 1, 2, ..., \eta \tag{20}$$

Note that in Eq. (20), K_p and B_p are irrelevant to the distance D_{ij}, and are determined only by $v_{\min}$, $v_{\max}$, and ε, which means the same set of K_p and B_p can be used for bounding the travel time of any arc with different lengths and travel speeds. Moreover, the same set of K_p and B_p can also be used for arcs with different lower/upper speed limits, i.e., $v_{\min}^{(i,j)}$ and $v_{\max}^{(i,j)}$, as long as $v_{\min}^{(i,j)} \ge v_{\min}$ and $v_{\max}^{(i,j)} \ge v_{\max}$ are satisfied. The values of K_p and B_p with respect to different ε for a speed range of [20,120] km/h are provided in Table 6. The reader may directly refer to these data to linearize the time and speed of their own VRP models.

From the table, the same inequalities as in Eq. (21) are used to bound e_3 against v for the third term $e_4(v) = \phi v^{-1}$, as follows.

$$e_4 \ge \phi K_p v + \phi B_p, \quad \forall p \in 1, 2, ..., \eta \tag{21}$$

where K_p and B_p are calculated as $K_p = -\mu^{2-2p}/v_{\min}^2$ and $B_p = 2\mu^{1-p}/v_{\min}$,

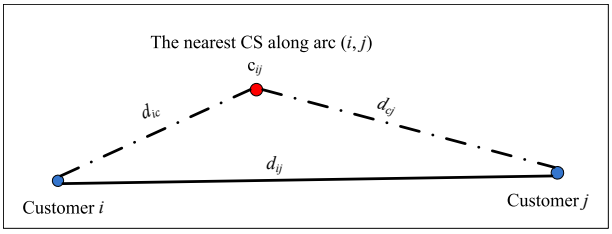


Fig. 9. An example of CS visit when traveling an arc.

Table 6
Data of Kp and Bp for speed range [20, 120] km/h.

ε	P	K_p	B_p	E	p	K_p	B_p
5.0 %	1	−2.50000E-03	1.00000E-01	0.1 %	1	−2.50000E-03	1.00000E-01
	2	−1.00651E-03	6.34512E-02		2	−2.20286E-03	9.38693E-02
	3	−4.05228E-04	4.02606E-02		3	−1.94104E-03	8.81145E-02
	4	−1.63147E-04	2.55458E-02		4	−1.71034E-03	8.27125E-02
	5	−6.56839E-05	1.62091E-02		5	−1.50706E-03	7.76416E-02
2.0 %	1	−2.50000E-03	1.00000E-01		6	−1.32793E-03	7.28817E-02
	2	−1.41452E-03	7.52201E-02		7	−1.17010E-03	6.84135E-02
	3	−8.00343E-04	5.65807E-02		8	−1.03103E-03	6.42193E-02
	4	−4.52840E-04	4.25601E-02		9	−9.08486E-04	6.02822E-02
	5	−2.56220E-04	3.20137E-02		10	−8.00508E-04	5.65865E-02
	6	−1.44971E-04	2.40808E-02		11	−7.05363E-04	5.31174E-02
	7	−8.20255E-05	1.81136E-02		12	−6.21527E-04	4.98609E-02
	8	−4.64106E-05	1.36251E-02		13	−5.47656E-04	4.68041E-02
1.0 %	1	−2.50000E-03	1.00000E-01		14	−4.82564E-04	4.39347E-02
	2	−1.67355E-03	8.18182E-02		15	−4.25209E-04	4.12412E-02
	3	−1.12031E-03	6.69422E-02		16	−3.74670E-04	3.87128E-02
	4	−7.49962E-04	5.47709E-02		17	−3.301389E-04	3.63395E-02
	5	−5.02040E-04	4.48125E-02		18	−2.90900E-04	3.41116E-02
	6	−3.36077E-04	3.66648E-02		19	−2.56325E-04	3.20203E-02
	7	−2.24977E-04	2.99985E-02		20	−2.25860E-04	3.00573E-02
	8	−1.50604E-04	2.45442E-02		21	−1.99015E-04	2.82145E-02
	9	−1.00818E-04	2.00816E-02		22	−1.75361E-04	2.64848E-02
	10	−6.74896E-05	1.64304E-02		23	−1.54519E-04	2.48611E-02
0.5 %	1	−2.50000E-03	1.00000E-01		24	−1.36153E-04	2.33369E-02
	2	−1.88321E-03	8.67918E-02		25	−1.19971E-04	2.19062E-02
	3	−1.41859E-03	7.53282E-02		26	−1.05712E-04	2.05632E-02
	4	−1.06859E-03	6.53787E-02		27	−9.31471E-05	1.93026E-02
	5	−8.04953E-04	5.67434E-02		28	−8.20761E-05	1.81192E-02
	6	−6.06357E-04	4.92486E-02		29	−7.23209E-05	1.70083E-02
	7	−4.56758E-04	4.27438E-02		30	−6.37252E-05	1.59656E-02
	8	−3.44067E-04	3.70981E-02				
	9	−2.59180E-04	3.21981E-02				
	10	−1.95235E-04	2.79453E-02				
	11	−1.47067E-04	2.42543E-02				
	12	−1.10783E-04	2.10507E-02				
	13	−8.34510E-05	1.82703E-02				
	14	−6.28621E-05	1.58571E-02				

$\mu = (1+\sqrt{\varepsilon})/(1-\sqrt{\varepsilon})$, and ε is the maximum allowable deviation from the actual value.

4. New formulation of CS visiting and recharging

It is generally supposed that the EV is permitted to visit one CS in the middle of the journey of traveling an arc (i, j). If a CS, c, is visited, then the total traversed distance from customers i to j is represented as $d_{ic} + d_{cj}$, where d_{ic} is the distance from i to c and d_{cj} is the distance from c to j; else, it is d_{ij}. However, if the distances between customers and CSs are known, the shortest distance for visiting a CS along the arc can always be known in advance. As shown in Fig. 9, let c_{ij} represent the nearest CS to arc (i, j). The total traversed distance for arc (i, j) is calculated as $d_{ij}(1 - y_{ij}) + D_{ij}y_{ij}$, where d_{ij} is the direct distance between customers/depot i and j, D_{ij} is the sum of detouring distances to the nearest CS c between customers/depot i and j, i.e., $D_{ij} = d_{ic} + d_{cj}$, and y_{ij} is a binary variable indicating whether the nearest CS is visited ($y_{ij} = 1$) or not ($y_{ij} = 1$) along the traveling arc (i, j). Notation E_{ij} is used to represent the energy consumed for traveling from customers i to j (the traveled distance can either be d_{ij} or D_{ij}) and notation $\widetilde{E}_{ij}$ is used to represent the energy consumed for traveling d_{ic} if the nearest CS is visited (Wang et al., 2013; Zhang et al., 2019) [77,78].

Zuo et al. (2019) [3] introduced four variables to describe the charging states of the battery of an EV that visits a CS, as follows.

p'_{ij} remaining energy of the EV battery when arriving at the nearest CS associated to arc (i, j)
p''_{ij} amount of energy recharged into the EV battery at the nearest CS associated to arc (i, j)
t'_{ij} charging time used to increase the EV battery energy from 0 to p'_{ij} at the nearest CS associated to arc (i, j)
t''_{ij} charging time used to increase the EV battery energy from p'_{ij} to $p'_{ij} + p''_{ij}$

The EV battery's state of charging (SOC) is a convex function, and the nonlinear relation between the SOC and the charging time can be represented by a number of secant lines on the SOC curve, as shown in Fig. 10, where (t_1, s_1), (t_2, s_2), …, (t_k, s_k), and (t_{k+1}, s_{k+1}) are the evenly distributed breakpoints and k is the number of lines. Thus, the introduced four variables satisfy the equations in Eq. (22) (Zuo et al., 2019) [3].

$$\begin{cases} p'_{ij} = k'_l t'_{ij} + b'_l \\ p'_{ij} + p''_{ij} \le k'_r\left(t'_{ij} + t''_{ij}\right) + b'_r \quad \forall r \in R \end{cases} \tag{22}$$

where k'_r and b'_r are slope and intercept of the line formed by breakpoints (t_r, s_r) and (t_{r+1}, s_{r+1}), $r \in R$, $l \in R$, R is the set of secant lines, and l satisfy $s_l \le p'_{ij} \le s_{l+1}$.

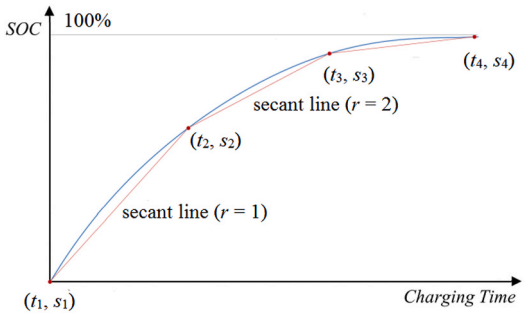


Fig. 10. A typical nonlinear charging function.

5. A new comprehensive model of EVRPTW

Based on the new formulations introduced in Sections 3 and 4, a new comprehensive formulation for the EVRPTW is presented in this section, with considerations of all the discussed EV characters. Notations used in the description of the model are listed in advance as follows.

Parameters:

[0, T] planning horizon
N set of all nodes including customers (positive integers) and the depot (zero)
N' set of customers without the depot.
i, j index of nodes, where $i, j \in N$
A set of customer arcs, where $i, j \in N$; $i \neq j$
f_i demand of customer i, $i \in N'$
c_{ij} the nearest CS associated to arc (i, j), $(i, j) \in A$
d_{ij} direct distance for traveling an arc (i, j), $(i, j) \in A$
D_{ij} summed distance from node i to the nearest CS c_{ij} and from the nearest CS c_{ij} to node j
$\widetilde{d}_{ij}$ distance from node i to the nearest CS c_{ij} associated to arc (i, j), $(i, j) \in A$
q number of homogeneous EVs available in the fleet
C battery capacity of EV (kWh)
O load capacity of EV
g_v fixed cost for per EV used
g_e unit price of commercial electricity (kWh)
g_d cost for one unit of traveled distance, e.g., rental, vehicle depreciation, tax, etc.
g_t cost for one unit of delivery time, e.g., driver's wage
$\underline{v}, \bar{v}$ lower/upper speed limit
ε maximum allowed linearization error rate
P set of tangential lines for the *time-speed* linearization
(K_r, B_r) slope and intercept of the rth line in P
Q set of tangential lines for $e_1(v) = \alpha v^2$ linearization
(k_r, k_r) slope and intercept of the rth line in Q
R set of consecutive secant lines for the SOC-time function linearization, and $k = \text{card}(R)$
(s_r, τ_r) the rth breakpoint in R, where s_r is the SOC (%), τ_r is the charging time, and $r = 1, 2, \ldots, k, k+1$.
(k'_r, b'_r) slope and intercept of the rth line in R
$[a_i, b_i]$ time window of customer i available for receiving the service, $a_i \in [0, T]$, $b_i \in [0, T]$
h_i service time needed for serving customer i
λ minimal remaining energy of an EV when arriving at CSs/depot
$\alpha, \beta, \gamma, \phi$, and χ coefficients of the ECR function in Eq. (13)

Binary decision variables:

xij indicating whether arc (i, j) is selected to traverse (xij = 1) or not (xij = 0)
yij indicating whether the EV detours to visit the nearest CS (yij = 1) or not (yij = 0) when traveling arc (i, j)
zijr indicating whether the remaining energy of an EV, namely, p'ij as defined bellow, when arriving at CS, cij, is in [sr, sr+1] (by zijr = 1) or not (by zijr = 0)

Non-negative continuous decision variables:

vij travel speed over arc (i, j)
tij travel time over arc (i, j)
e1ij first term of ECR in Eq. (14) over arc (i, j)
e4ij fourth term of ECR in Eq. (14) over arc (i, j)
Eij energy consumed over arc (i, j) in kWh
$\widetilde{E}_{ij}$ energy consumed from node i to the nearest CS cij associated to arc (i, j) in kWh

ui start time to serve customer i
pi remaining energy of the EV when it arrives at customer i
p'ij remaining energy of the EV when it at the nearest CS, i.e., cij, to arc (i, j)
p"ij recharged energy at the CS cij
t'ij time consumed to charge the energy of a drained EV from zero to p'ij
t"ij time consumed to charge the energy of a non-drained EV from p'ij to p'ij + p"ij
lij payload of an EV when traveling arc (i, j)
wi waiting time between the arrival time and service starting time of customer i
w'i waiting time between the service completing time and leaving time of customer i

The objective is to minimize the total cost of employing/owning a fleet of EVs to visit a set of customers, including vehicles' fixed cost, drivers' wage, electricity cost, and travel cost. The objective function is expressed in Eq. (23), as follows.

$$\begin{aligned}\text{Minimize Total Cost} = {} & g_v \sum_{(0,j)\in A} x_{0j} \\ & + g_e \sum_{(i,j)\in A} E_{ij} \\ & + g_d \sum_{(i,j)\in A} \left[x_{ij} d_{ij} + y_{ij}\left(D_{ij} - d_{ij}\right)\right] \\ & + g_t \left[\sum_{(i,j)\in A}\left(t_{ij} + t''_{ij}\right) + \sum_{i\in N'}\left(w_i + h_i + w'_i\right)\right]\end{aligned} \tag{23}$$

The above objective function and variables subject to eight groups of linear constraints expressed in Eqs. (24) – (35), as follows.

(1) Constraints to guarantee feasible visiting routes of all EVs.

$$\sum_{(j,i)\in A} x_{ji} = 1 \quad \forall i \in N' \tag{24}$$

$$\sum_{(i,j)\in A} x_{ij} = 1 \quad \forall i \in N' \tag{25}$$

$$\sum_{i\in N'} x_{0i} \le q \tag{26}$$

(2) Constraints to guarantee the linearized time-speed relation deduced in Section 3.2.2.

$$y_{ij} \le x_{ij} \quad (i,j) \in A \tag{27}$$

$$\underline{v} x_{ij} \le v_{ij} \le \bar{v} x_{ij} \quad (i,j) \in A \tag{28}$$

$$\begin{cases} t_{ij} \ge d_{ij} K_r v_{ij} + d_{ij} B_r - T\left(1 - x_{ij} + y_{ij}\right) & \forall (i,j) \in A, r \in P \\ t_{ij} \ge D_{ij} K_r v_{ij} + D_{ij} B_r - T\left(2 - x_{ij} - y_{ij}\right) & \forall (i,j) \in A, r \in P \end{cases} \tag{29}$$

(3) Constraints to bound the arrival time of EVs at all customers.

$$\begin{cases} u_i \ge t_{0i} + w_i - T(1 - x_{0i}) & \forall (0,i) \in A & (30\text{-}1) \\ u_i \le t_{0i} + w_i + T(1 - x_{0i}) & \forall (0,i) \in A & (30\text{-}2) \\ u_j - u_i \ge h_i + w'_i + t_{ij} + t''_{ij} + w_j - T(1 - x_{ij}) & \forall (i,j) \in A, i > 0, j > 0 & (30\text{-}3) \\ u_j - u_i \le h_i + w'_i + t_{ij} + t''_{ij} + w_j + T(1 - x_{ij}) & \forall (i,j) \in A, i > 0, j > 0 & (30\text{-}4) \\ a_i \le u_i \le b_i & \forall i \in N' & (30\text{-}5) \end{cases}$$

(4) Constraints to determine the payloads of EVs on the traversed arcs along their tours.

$$\begin{cases} \sum_{(j,i)\in A} l_{ji} - \sum_{(i,j)\in A} l_{ij} = f_i & \forall i \in N' & (31\text{-}1) \\ l_{ij} \le O x_{ij} & \forall (i,j) \in A & (31\text{-}2) \end{cases}$$

(5) Constraints to calculate the amounts of energy consumption on all traveled arcs according to the linearized ECR function deduced in Section 3.2.

$$\begin{cases} e_{1ij} \ge \alpha k_r v_{ij} + \alpha b_r - C(1 - x_{ij}) & \forall (i,j) \in A, r \in P & (32\text{-}1) \\ e_{4ij} \ge \varphi K_r v_{ij} + \varphi B_r - C(1 - x_{ij}) & \forall (i,j) \in A, r \in P & (32\text{-}2) \\ E_{ij} \ge (e_{1ij} + \beta v_{ij} + \gamma + e_{4ij} + \chi f_{ij}) d_{ij} - C(1 - x_{ij} + y_{ij}) & \forall (i,j) \in A & (32\text{-}3) \\ E_{ij} \ge \tilde{E}_{ij} + (e_{1ij} + \beta v_{ij} + \gamma + e_{4ij} + \chi f_{ij})(D_{ij} - \tilde{d}_{ij}) - C(2 - x_{ij} - y_{ij}) & \forall (i,j) \in A & (32\text{-}4) \\ \tilde{E}_{ij} \ge (e_{1ij} + \beta v_{ij} + \gamma + e_{4ij} + \chi f_{ij}) \tilde{d}_{ij} - C(2 - x_{ij} - y_{ij}) & \forall (i,j) \in A & (32\text{-}5) \end{cases}$$

(6) Constraints to determine the amounts of remaining energy of EV at all arrival nodes based on the CS visiting formulation introduced in Section 4.

$$\begin{cases} C - p_i \ge E_{0i} - C(1 - x_{0i}) & \forall (0,i) \in A & (33\text{-}1) \\ C - p_i \le E_{0i} + C(1 - x_{0i}) & \forall (0,i) \in A & (33\text{-}2) \\ p_i + p''_{ij} - p_j \ge E_{ij} - C(1 - x_{ij}) & \forall (i,j) \in A, i > 0, j > 0 & (33\text{-}3) \\ p_i + p''_{ij} - p_j \le E_{ij} + C(1 - x_{ij}) & \forall (i,j) \in A, i > 0, j > 0 & (33\text{-}4) \\ p_i + p''_{i0} \ge E_{i0} + \lambda - C(1 - x_{i0}) & \forall (i,0) \in A & (33\text{-}5) \\ p_i - p'_{ij} \ge \tilde{E}_{ij} - C(1 - y_{ij}) & \forall (i,j) \in A & (33\text{-}6) \\ p_i - p'_{ij} \le \tilde{E}_{ij} + C(1 - y_{ij}) & \forall (i,j) \in A & (33\text{-}7) \\ p'_{ij} \ge \lambda y_{ij} & \forall (i,j) \in A & (33\text{-}8) \end{cases}$$

(7) Constraints to re-charge the EVs by the linearization method introduced by Zuo et al. (2019) and repeated in Section 4.

$$\begin{cases} s_r + (z_{ijr} - 1) \le p'_{ij} / C \le s_{r+1} + (1 - z_{ijr}) & \forall (i,j) \in A, r \in R & (34\text{-}1) \\ \sum_{r\in R} z_{ijr} = y_{ij} & \forall (i,j) \in A & (34\text{-}2) \\ p'_{ij} / C \ge k'_r t'_{ij} + b'_r - (1 - z_{ijr}) & \forall (i,j) \in A, r \in R & (34\text{-}3) \\ p'_{ij} / C \le k'_r t'_{ij} + b'_r + (1 - z_{ijr}) & \forall (i,j) \in A, r \in R & (34\text{-}4) \\ (p'_{ij} + p''_{ij}) / C \le k'_r (t'_{ij} + t''_{ij}) + b'_r + (1 - y_{ij}) & \forall (i,j) \in A, r \in R & (34\text{-}5) \\ p'_{ij} + p''_{ij} \le C y_{ij} & \forall (i,j) \in A & (34\text{-}6) \\ t'_{ij} + t''_{ij} \le T y_{ij} & \forall (i,j) \in A & (34\text{-}7) \end{cases}$$

(8) Constraints to define the value domains for all variables.

$$\begin{cases} x_{ij}, y_{ij}, \alpha_{ijr} \in \{0,1\}; v_{ij} \geq 0; t_{ij} \geq 0; u_i \geq 0; p_i \geq 0; p'_{ij} \geq 0; p''_{ij} \geq 0; \\ e_{1ij} \geq 0; e_{2ij} \geq 0; t'_{ij} \geq 0; t''_{ij} \geq 0; l_{ij} \geq 0; w_i \geq 0; w'_i \geq 0; E_{ij} \geq 0; \widetilde{E}_{ij} \geq 0 \end{cases} \quad \forall i,j,c,r \tag{35}$$

The above formulas are explained in more details in this paragraph. The objective in Eq. (23) is a comprehensive cost function including four parts: (i) sum of fixed cost of all used vehicles, such as vehicle's rental and/or driver's fixed wage; (ii) energy cost paid for consumed electricity; (iii) cost related to the traveled distance, such as driver's flexible wage (if applicable) and/or vehicle mileage-based depreciation; and (iv) other cost related to the delivering time (if applicable). Constraints (24) and (25) guarantees that each customer is visited exactly once. Constraint (26) restricts the total number of available EVs. Constraint (27) indicates the nearest CS to arc (i, j) can be visited only when the arc is selected for travel. Constraint (28) restricts the speed variable between the lower and upper limits. Constraint (29-1) bounds the travel time of traveled arcs where CSs are not visited (i.e., $x_{ij} = 1$ and $y_{ij} = 0$). Constraint (29-2) bounds the travel time on the traversed arcs where CSs are visited (i.e., $x_{ij} = 1$ and $y_{ij} = 1$). Constraints (30-1)–(30-5) restrict the time flow, e.g., traveling time, arrival time, waiting time, and service time, along the route and guarantee that all customers' time window requirement are satisfied. Constraints (31-1)–(31-2) balance the load flow as the EV delivering (or collecting) goods along the route and enforce a load capacity limit to all EVs. Constraints (32-1)–(32-5) determine the energy consumption for traveled arcs, where Constraints (32-1) and (32-2) bound the first and second parts of the ECR function in Eq. (15), respectively. Constraint (32-3) calculates the energy consumed for traveling the arcs that do not visit a CS in the middle. Constraint (32-4) calculates the energy consumed for traveling the arcs where a CS (the nearest one) must be visited in the middle. Constraint (32-5) calculates the energy consumed for visiting the nearest CS. Constraints (33-1)–(33-8) restrict the energy flow balance, i.e., initial charge, consumption and recharging, along the route, where the first arc and last arc of a route are determined by Constraints (33-1)-(33-2) and Constraint (33-5), respectively, the rest arcs are determined uniformly by Constraints (33-3) and (33-4), Constraints (33-6) and (33-7) determine the remaining energy of an EV, i.e., p'_{ij}, when it arrives the nearest CS of arc (i, j) (if applicable by $y_{ij} = 1$). The residual energy of an EV is required to be above a given safety threshold, λ, when it arrives at an CS (by Constraint (33-8) or backs to the depot (by Constraint (33-5)).

Constraints (34-1)–(34-7) are a set of linear inequalities that guarantees the implementation of the partial charging policy with flexible charging time for an EV having a concave nonlinear charging function (Zuo et al., 2019) [3]. Constraints (34-1) and (34-2) choose a secant line, r, $r \in R$, to represent the remaining energy of an EV when it arrives the nearest CS of arc (i, j). Constraints (34-3) and (34-4) bound variable t'_{ij} to p'_i by using the chosen line. Constraint (34-5) bounds the charged energy $(p'_{ij} + p''_{ij})$ to the charging time $(t'_{ij} + t''_{ij})$. The maximum battery capacity of EV is guaranteed not to exceed by Constraint (34-6), and the total charging time is guaranteed not to exceed the planning horizon by Constraint (34-7). The value domains of all decision variables are defined by Constraint (35).

6. Computational experiments

The newly developed comprehensive model of EVRP in Section 5 is linear and it is solvable to many off-shelf optimization tools such as CPLEX and Lingo. In this section, the solver CPLEX (version 12.6.0.1) was employed to solve the test instances. The model was coded by AMPL (A Mathematical Programming Language) and solved by CPLEX on a linux PC server with two CPUs (32 threads) of 2.9 GHz Intel Xeon (R). The physical memory is 128 G RAM.

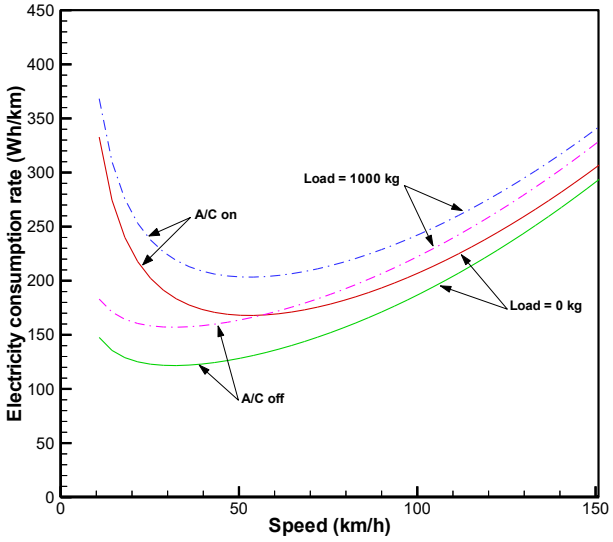


Fig. 11. Change in ECR with respect to speed under different conditions.

6.1. Test instances

Three sets of test instances were generated based on the first group of the well-known Solomon's VRPTW benchmark instances (Solomon, 1987) [79]. The test data set includes 12 instances of R1-type (randomly generated customer locations), 9 instances of C1-type (clustered customer locations), and 8 instances of RC1-type (mixture of random and clustered customer location). Each instance contains one central depot and 100 customers with specific location coordinates with Euclidean distance matric, demands for delivering/collecting, service time duration, and time windows in which the service must start.

The distance unit is km, the time unit is min, the speed unit is km/h, the load unit is kg, and the electricityunit is W-h. A fleet of homogeneous EVs with ECR Parameters listed in Table 3 in Section 3.1 are going to serve these customers. The EVs are fully charged when departing from the depot, consume their batteries along the routes, and maintain the remaining energies being above a given safe level (i.e., 15 %). The ECR function (Eq. (7)) was linearized by summing up all electricity consumptions resulting from aerodynamic, tire, drivetrain, ancillary, and kinetic/potential losses. The cost settings for the vehicle/driver were taken from Xiao et al. (2019) [39]. The number of available EVs in the fleet was set to 5. The EVs was set with a load capacity of 200 units.

For each Solomon's instance, the depot and the first 11 customers were chosen to form a 11-customer instance for testing, i.e., $|N| = 12$. Accordingly, four CSs were set in four angular directions of the customer map. They are available for all EVs to recharge. The x-coordinates of the CSs were calculated by $\min\{x_i\} + 0.25\max\{|x_i - x_j|\}$ or $\min\{x_i\} + 0.75\max\{|x_i - x_j|\}$, and the y-coordinates were set by $\min\{y_i\} + 0.25\max\{|y_i - y_j|\}$ or $\min\{y_i\} + 0.75\max\{|y_i - y_j|\}$, where i and j represent nodes. In all experiments, the accuracy parameter ε for the linearization of the ECR model and for the travel time linearization of $t_{ij} = d_{ij}/v_{ij}$ was set as $\varepsilon = 1$ %. If the calculation cannot be completed with a time limit of 7200 s, then the best feasible solution found so far is taken as the outcome solution.

6.2. Experiments on the effects of A/C and load

According to the ECR proposed in Section 3, the ECR curves of EV is

Table 7
Experiments on 29 of 11-customer instances with A/C on and off ($n = 12$).

Problem Ins.		A/C ON								A/C OFF							
Type	ID	Obj.	V.	Dis.	Total T.	Total E.	CPU T.	Exact	Dev.	Obj.	V.	Dis	Total T.	Total E.	CPU T.	Exact	Dev.
		(¥)		(km)	(min)	(kW·h)	(s)	(¥)	(%)	(¥)		(km)	(min)	(kW·h)	(s)	(¥)	(%)
R1-type	R101	**1921.84**	4	297.8	783.4	71.5	7	**1923.42**	−0.08	**1895.80**	4	297.8	813.5	64.3	8	**1897.72**	−0.10
	R102	**1573.67**	3	269.8	716.0	69.5	1213	**1575.15**	−0.09	**1551.27**	3	269.1	741.0	63.6	562	**1552.52**	−0.08
	R103	**1572.14**	3	269.1	716.0	69.4	1173	**1573.55**	−0.09	**1551.27**	3	269.1	741.0	63.6	212	**1552.52**	−0.08
	R104	**527.82**	3	58.4	1146.7	10.8	55	**528.21**	−0.07	**521.67**	3	58.4	1156.9	9.0	16	**522.10**	−0.08
	R105	**1873.04**	4	291.6	788.4	61.1	172	**1874.76**	−0.09	**1846.78**	4	291.6	822.0	53.7	52	**1848.64**	−0.10
	R106	**1501.73**	3	250.3	631.2	59.6	5906	**1503.23**	−0.10	**1479.63**	3	249.7	645.6	53.9	2182	**1481.20**	−0.11
	R107	**1501.73**	3	250.3	631.2	59.6	5386	**1503.23**	−0.10	**1479.63**	3	249.7	645.6	53.9	778	**1481.20**	−0.11
	R108	**530.08**	1	59.5	1150.6	10.9	8	**530.47**	−0.07	**523.53**	1	59.5	1168.5	8.8	3	**524.04**	−0.10
	R109	**1519.54**	3	259.8	710.7	59.2	395	**1520.80**	−0.08	**1496.45**	3	259.8	735.4	52.8	45	**1497.80**	−0.09
	R110	**1465.47**	3	232.6	585.6	56.9	3972	**1467.10**	−0.11	**1446.69**	3	232.0	596.7	52.1	675	**1446.18**	0.04
	R111	**1494.42**	3	250.3	642.2	57.5	7107	**1495.84**	−0.10	**1471.45**	3	249.7	656.6	51.5	1162	**1472.75**	−0.09
	R112	1179.85	3	219.2	543.8	65.6	7200	1181.55	−0.14	1142.06	3	219.0	547.4	56.1	7200	1142.18	−0.01
	AVG	1388.32	3	225.7	753.8	54.29	2716	1389.64	−0.09	1367.19	3	225.4	772.5	48.60	1800	1368.24	−0.08
C1-type	C101	**540.82**	1	59.5	1105.8	14.7	1	**541.25**	−0.08	536.32	1	59.5	1112.0	13.4	2	**536.76**	−0.08
	C102	**527.97**	1	58.4	1144.4	10.9	87	**528.35**	−0.07	521.88	1	58.4	1155.7	9.1	18	**522.31**	−0.08
	C103	**527.98**	1	58.4	1144.4	10.9	83	**528.35**	−0.07	521.88	1	58.4	1155.7	9.1	23	**522.30**	−0.08
	C104	**527.82**	1	58.4	1146.7	10.8	156	**528.19**	−0.07	521.67	1	58.4	1156.9	9.0	53	**522.09**	−0.08
	C105	**531.99**	1	59.5	1128.8	11.9	1	**532.35**	−0.07	526.64	1	59.5	1135.0	10.4	2	**527.02**	−0.07
	C106	**535.71**	1	59.5	1114.8	13.2	1	**536.10**	−0.07	530.88	1	59.5	1121.0	11.8	1	**531.28**	−0.08
	C107	**530.08**	1	59.5	1150.6	10.9	1	**530.47**	−0.07	523.53	1	59.5	1168.5	8.8	1	**524.04**	−0.10
	C108	**530.08**	1	59.5	1150.6	10.9	9	**530.47**	−0.07	523.53	1	59.5	1168.5	8.8	12	**524.04**	−0.10
	C109	**530.07**	1	59.5	1150.5	10.9	54	**530.47**	−0.08	523.53	1	59.5	1168.5	8.8	18	**524.04**	−0.10
	AVG	531.39	1	59.13	1137.4	11.67	44	531.78	−0.07	525.54	1	59.13	1149.1	9.92	15	525.99	−0.09
RC1-type	RC101	**1459.83**	3	244.7	695.2	49.2	93	**1461.00**	−0.08	1436.97	3	244.7	730.1	42.6	22	**1438.33**	−0.09
	RC102	**1002.68**	2	175.2	536.2	34.7	107	**1003.58**	−0.09	986.08	2	175.2	560.3	29.9	33	**987.18**	−0.11
	RC103	**1002.68**	2	175.2	536.2	34.7	114	**1003.63**	−0.09	986.08	2	175.2	560.3	29.9	35	**987.18**	−0.11
	RC104	**527.82**	1	58.4	1146.7	10.8	16	**528.21**	−0.07	521.67	1	58.4	1156.9	9.0	6	**522.10**	−0.08
	RC105	**1026.05**	2	180.7	526.2	39.1	78	**1026.92**	−0.08	1009.82	2	180.7	550.3	34.5	13	**1010.86**	−0.10
	RC106	**1036.17**	2	180.6	501.2	42.3	118	**1037.27**	−0.11	1020.51	2	180.5	522.1	37.9	17	**1021.51**	−0.10
	RC107	**1002.98**	2	170.6	513.4	36.7	192	**1004.20**	−0.12	986.97	2	170.6	545.0	32.0	164	**988.42**	−0.15
	RC108	**991.96**	2	170.6	526.2	33.7	187	**993.01**	−0.11	975.44	2	170.6	558.8	28.7	127	**976.56**	−0.11
	AVG	1006.27	2	169.5	622.7	35.15	113	1007.23	−0.09	990.44	2	169.5	648.0	30.57	52	991.52	−0.11

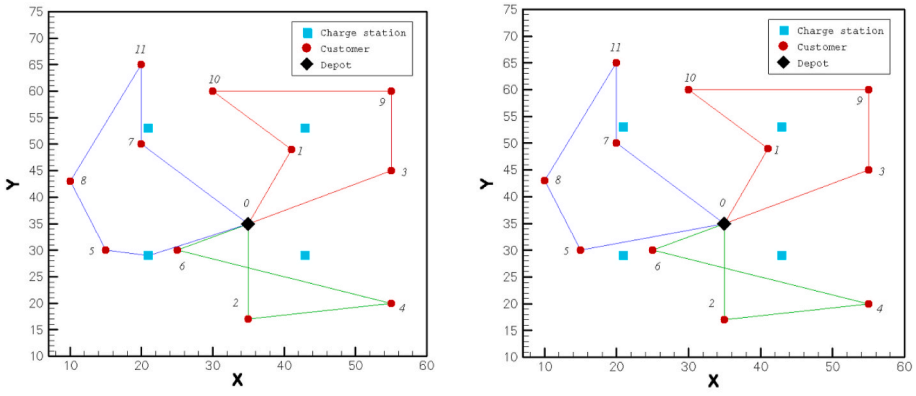
Note: Bold face indicates optimal values.

Table 8
Detailed schedules of the solution for instance R-102 with 11 customers (Obj. = 1573.67, $n = 12$, A/C on).

Veh. No.	Arc (i,j)	D_{ij}	v_{ij}	t_{ij}	t'_{ij}	w_i	u_i	$[h_i,e_i]$	w'_i	E_{ij}	E'_{ij}	a_i	l_{ij}	p_i	p''_{ij}	t''_{ij}
		(km)	(km/h)	(min)	(min)	(min)	(min)	(min)	(min)	(W-h)	(W-h)	(kg)	(kg)	(W-h)	(W)	(min)
1	(0,3)	22.4	26.9	49.4	49.9	0.0	49.4	[0,197]	0.0	4544.6	4566.3	13.0	55.0	25,455.4	0.0	0.0
	(3,9)	15.0	23.9	37.6	37.7	0.0	97.0	[97,107]	0.0	2854.1	2853.4	16.0	42.0	22,601.3	0.0	0.0
	(9,10)	25.0	55.5	27	27.0	0.0	134.0	[124,134]	0.0	11,227.3	11218.9	16.0	26.0	11,374.0	0.0	0.0
	(10,1)	15.6	22.0	42	42.5	0.0	204.0	[0,204]	0.0	2816.2	2831.3	10.0	10.0	8557.8	0.0	0.0
	(1,0)	15.2	22.0	41.1	41.5	–	–	[0,230]	–	2752.0	2766.8	0.0	0.0	805.8	0.0	0.0
2	(0,6)	11.2	20.0	33	33.6	0.0	99.0	[99,109]	0.0	1950.3	1974.4	3.0	29.0	28,049.7	0.0	0.0
	(6,4)	31.6	37.9	50	50.0	0.0	159.0	[149,159]	0.0	8658.1	8667.2	19.0	26.0	19,391.6	0.0	0.0
	(4,2)	20.2	36.8	33	32.9	0.0	202.0	[0,202]	0.0	5352.8	5357.7	7.0	7.0	14,038.8	0.0	0.0
	(2,0)	18.0	22.0	48.6	49.1	–	–	[0,230]	–	3252.2	3269.7	0.0	0.0	10,786.6	0.0	0.0
3	(0,7)	21.2	40.2	31.4	31.6	0.0	31.4	[0,198]	0.0	6189.7	6230.2	5.0	52.0	23,810.3	0.0	0.0
	(7,11)	15.0	35.1	25.6	25.6	0.0	67.0	[67,77]	0.0	3799.0	3807.3	12.0	47.0	20,011.3	0.0	0.0
	(11,8)	24.2	51.6	28	28.1	0.0	105.0	[95,105]	0.0	9754.3	9767.7	9.0	35.0	10,257.0	0.0	0.0
	(8,5)	13.9	22.0	37.6	37.9	0.0	152.6	[0,199]	0.0	2529.4	2542.9	26.0	26.0	7727.7	0.0	0.0
	(5,0)	21.3	22.0	57.6	58.1	–	–	[0,230]	–	3851.0	3869.2	0.0	0.0	5000.0	1123.3	1.2
Total		269.8		541.9	545.6	0.0			0.0	69,530.9	69723.0				1123.3	1.2

Table 9
Detailed schedules of the solution for instance R-102 with 11 customers (Obj. = 1551.27, $n = 12$, A/C off).

Veh. No	Arc (i,j)	D_{ij}	v_{ij}	t_{ij}	t'_{ij}	w_i	u_i	$[h_i,e_i]$	w'_i	E_{ij}	E'_{ij}	a_i	l_{ij}	p_i	p''_{ij}	t''_{ij}
		(km)	(km/h)	(min)	(min)	(min)	(min)	(min)	(min)	(W-h)	(W-h)	(kg)	(kg)	(W-h)	(W)	(min)
1	(0,3)	22.4	26.9	49.4	49.9	0.0	49.4	[0,197]	0.0	4087.2	4103.1	13.0	55.0	25,912.8	0.0	0.0
	(3,9)	15.0	23.9	37.6	37.6	0.0	97.0	[97,107]	0.0	2505.9	2506.4	16.0	42.0	23,406.8	0.0	0.0
	(9,10)	25.0	55.5	27.0	27.0	0.0	134.0	[124,134]	0.0	10,977.3	10,980.0	16.0	26.0	12,429.5	0.0	0.0
	(10,1)	15.6	20.0	46.7	46.7	0.0	190.7	[0,204]	0.0	2271.1	2304.6	10.0	10.0	10,158.4	0.0	0.0
	(1,0)	15.2	20.0	45.7	45.7	–	–	[0,230]	–	2218.3	2251.1	0.0	0.0	7940.2	0.0	0.0
2	(0,6)	11.2	20.0	33.5	33.5	65.5	99.0	[99,109]	0.0	1639.8	1663.9	3.0	29.0	28,360.2	0.0	0.0
	(6,4)	31.6	37.9	50.0	50.1	0.0	159.0	[149,159]	0.0	8195.2	8201.9	19.0	26.0	20,165.1	0.0	0.0
	(4,2)	20.2	36.8	33.0	33.0	0.0	202.0	[0,202]	0.0	5047.2	5047.4	7.0	7.0	15,117.9	0.0	0.0
	(2,0)	18.0	20.0	54.0	54.0	–	–	[0,230]	–	2621.5	2660.3	0.0	0.0	12,496.4	0.0	0.0
3	(0,7)	21.2	40.2	31.4	31.7	0.0	31.4	[0,198]	0.0	5899.3	5930.9	5.0	52.0	24,100.7	0.0	0.0
	(7,11)	15.0	35.1	25.6	25.7	0.0	67.0	[67,77]	0.0	3561.7	3565.2	12.0	47.0	20,539.1	0.0	0.0
	(11,8)	24.2	51.6	28.0	28.1	0.0	105.0	[95,105]	0.0	9495.0	9517.6	9.0	35.0	11,044.1	0.0	0.0
	(8,5)	13.9	20.0	41.8	41.8	0.0	199.0	[0,199]	0.0	2041.3	2071.4	26.0	26.0	9002.8	0.0	0.0
	(5,0)	20.6	20.0	61.9	61.8	–	–	[0,230]	–	3002.4	3046.8	0.0	0.0	6000.4	0.0	0.0
Total		269.1		565.5	566.5	65.5				63,563.1	63,850.6				0.0	0.0



(a) With A/C on (Total cost = 1,573.78) (b) With A/C off (Total cost = 1,551.27)

Fig. 12. Routing schemes obtained for the R102 instance.

plotted in Fig. 11 with respect to different loads (with load 1000 kg or no load) and states of the air condition (A/C on or off).

In all four cases in Fig. 11, ECR tends to fall with the increase of speed, before rising again. However, at low speeds, the value of ECR with the A/C turned on is significantly different from that with the A/C turned off. Take the solid green and the solid red curves as examples (i.e., load = 0). At the speed of 10 km/h, the value of ECR with A/C turned on is more than twice of that with A/C turned off (330 W-h/km and 150 W-h/km, respectively). Subsequently, with the increase of speed, the ECR with A/C turned on reaches its lowest point of 120 W-h/km at 30 km/h, while with A/C turned off, it reaches its lowest point of 170 W-h/km at 50 km/h. Finally, the two curves rise steadily with speed. To sum up, the economic speed of EVs is approximately 30 km/h with the A/C turned on and 50 km/h with the A/C turned off.

Next, all 29 instances were solved with the A/C turned on and turned off, respectively. The battery capacity was set to 30 kW h. The results are

compared in Table 7 with A/C on and off, where columns *Obj*., *V.*, *Dis.*, *Total T.*, *Total E.*, and *CPU T.* represent respectively objective values of the obtained solutions, number of vehicles used, total distance traveled by all EVs, total delivery time of all EVs (traveling time + waiting time + service time), total energy consumed by all EVs, and CPU time used in seconds.. Column *Exact* represent the exact solutions that were obtained based on the solutions in Column *Obj.*, where the amounts of energy consumed on each arc were re-calculated by using E.q. (9) and the travel times were re-calculated by using d_{ij}/v_{ij}. Column Dev. represents the deviations of the solutions in Column *Obj.* from those in Column *Exact*.

As shown in Table 7, the battery power consumes much faster when the A/C is turned on, resulting in more detours to visit CSs. For each of the 29 tested problem instances, the total consumed energy with the A/C turned off is −13.31 % lower than those with A/C turned on on average. For the instances R108, C107, C108, and C109 in particular, the reduction rates are nearly −20 %. Accordingly, the minimum objective function values of each instance with the A/C turned off are lower than those with the A/C turned on. The average deviation of the approximated solutions by the linearization techniques are about −0.1 % from the exact solutions.

In Tables 8 and 9, two solutions of instance R102 with the A/C turned on and turned off, respectively, were provided in details as a contrast for readers to understand the scheduling details. The routing schemes of the two solutions are plotted in Fig. 12. As observed, in the solution all the arcs were traveled with different but optimized speeds. Because the economic speed with the A/C turned on is higher than that with the A/C turned off (Fig. 11), the travel time with the A/C turned off was longer than that with the A/C turned on, such as the arcs (10,1), (1,0), (8,5) and (5,0). In Tables 8 and 9, Column t_{ij} and E_{ij} represent the approximated travel time and the consumed electricity energy according to the linearization of the ECR model. While Columns t'_{ij} and E'_{ij} provide the exact values for comparisons, where Column t'_{ij} was calculated by $t'_{ij} = D_{ij}/v_{ij}$ and Column E'_{ij} was calculated by the ECR formula in Eq. (9) timed by D_{ij}. The approximated solutions for instance R-102 are 1573.67 (A/C on) and 1551.27 (A/C off), which are about −0.08 % and −0.10 % deviated from the exact solutions 1575.15 (A/C on) and 1552.52 (A/C off), respectively.

6.3. Experiments on the effect of battery capacity

In this section, the effect of battery capacity is tested based on the 8 RC1-type instances. The experiments were conducted with changing battery capacities while other experimental conditions were the same with those of Section 6.2. The battery of EVs was set to take capacities of 16, 20, 24, and 32 kW-h (with the same the fixed cost), respectively, and the instances were solved in CPLEX with the optimal solutions. The computational results are provided and compared in Tables 10 and 11, and Fig. 13.

From Table 10, it is observed that the variations of battery capacity have affected considerably on the transportation costs. A higher battery capacity setting, i.e., 32 kW-h, has always resulted in solutions with lower objective values (total costs) than those obtained by lower battery capacity settings, with average cost deviations of 21.23 %, 9.73 %, 1.70 %, and 0.19 % corresponding to the battery capacities of 16, 20, and 24 kW-h, respectively. This is because a battery with a lower capacity often prompts the driver to take detours to CSs, consequently increasing the total distance and total time. In addition, since EVs with a lower battery capacity have a shorter driving range, more EVs will be required to complete the deliveries, resulting in higher fixed costs.

The average performances are summarized in Table 11, where the total distance, total time, total energy, and total cost are listed against different better capacity settings. The total cost includes the fixed cost of used EVs, cost related to distance, cost related to time, and energy cost. As observed that the average total cost for a low capacity setting C = 16 is the highest one, i.e., 1277.55. While when the capacity setting was doubled, i.e., C = 32, the average total cost dropped to 1006.28. The major difference is mainly on the fixed cost part, with a deviation of 31.25 %. The distance cost and time cost have respectively deviations of 23.80 % and 27.22 % in these calculations. However, they may become major differences in some countries where the drivers' wage rates are high. The energy cost has the lowest deviation (13.10 %) because the unit price setting of electricity is relatively lower than others. It is also notable that in this experiment the fixed cost of an EV was set to be independent of its batter capacity. This might not be true in reality because an EV with a larger capacity could be more expensive and might result in a higher fixed cost. So the EVs may not need necessarily to have a very larger battery capacity in real logistic systems.

Fig. 13 (a) shows the trends of variable cost, which consists of distance cost, time cost, and energy cost. Fig. 13 (b) illustrates the trends of the total cost, which consists of variable cost and fixed cost. In general, all costs decrease steadily as the battery capacity increases from 16 to 32 kW h, and remain stable thereafter.

6.4. Experiments on the effect of travel speed

In this section, the performance of the proposed comprehensive model for the EVRPTW was tested by the C1-type instances. First, the nine instances were solved by setting the travel speed (on each arc) as continuous variables. Next, they were solved with fixed travel speeds (for all arcs) of 20, 40, 60, 80, and 100 km/h. The total costs of solutions obtained under different speed settings are shown in Tables 12 and 13, and Fig. 14.

From Table 12, variations in speed significantly influence the transportation cost. The solutions obtained at different fixed travel speeds always have higher objective values (total costs) than those obtained at flexible travel speeds. The average cost increase rates can be

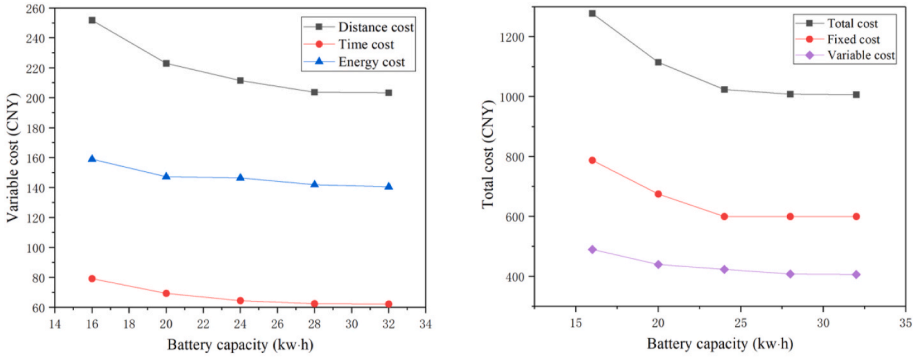
Table 10
Solutions obtained under different battery capacities with a safe threshold of 15 % (A/C on).

Problem Ins.		Different battery capacities (kW-h)									
Type	ID	16		20		24		28		32	
		Total_cost (¥)	Dev. (%)	Total_cost (¥)	Dev. (%)	Total_cost (¥)	Dev. (%)	Total_cost (¥)	Dev. (%)	Total_cost (¥)	Dev. (%)
RC1-type	RC101	1476.55	1.13	1470.48	0.72	**1459.83**	0.00	**1459.83**	0.00	**1459.83**	0.00
	RC102	1440.01	30.37	1003.82	0.11	1003.11	0.04	**1002.68**	0.00	**1002.68**	0.00
	RC103	1440.38	30.39	1003.82	0.11	1003.11	0.04	**1002.68**	0.00	**1002.68**	0.00
	RC104	**527.82**	0.00	**527.82**	0.00	**527.82**	0.00	**527.82**	0.00	**527.82**	0.00
	RC105	1446.35	29.06	1048.29	2.12	1032.81	0.65	1026.54	0.05	**1026.05**	0.00
	RC106	1444.93	28.29	1438.70	27.98	1162.64	10.88	1051.15	1.42	**1036.20**	0.00
	RC107	1434.88	30.10	1431.77	29.95	1008.25	0.52	**1002.98**	0.00	**1002.98**	0.00
	RC108	1009.54	1.74	993.56	0.16	**991.96**	0.00	**991.96**	0.00	**991.96**	0.00
	AVG	1315.85	21.23	1114.78	9.73	1023.69	1.70	1008.21	0.19	1006.28	0.00

Note: Bold face indicates optimal values.

Table 11
Solutions for RC-type instances under different battery capacities with the safe threshold of 15 % (A/C on).

Battery capacity	Average base on 8 RC1-type instances							
	Total distance	Total time	Total energy	Total cost	Fixed cost	Distance cost	Time cost	Energy cost
(kW-h)	(km)	(min)	(min)	(¥)	(¥)	(¥)	(¥)	(¥)
$C = 16$	209.84	792.23	39,757.24	1277.55	787.50	251.80	79.22	159.03
$C = 20$	185.86	694.20	36,831.11	1114.78	675.00	223.04	69.42	147.32
$C = 24$	176.32	645.53	36,887.76	1022.68	600.00	211.58	64.55	146.55
$C = 28$	169.78	624.88	35,493.93	1008.21	600.00	203.74	62.49	141.98
$C = 32$	169.49	622.72	35,149.29	1006.28	600.00	203.39	62.27	140.60



(a) Variable cost variation trends (b) Total cost variation trends

Fig. 13. Trends of cost variation with respect to different battery capacities.

Table 12
Solutions obtained at flexible and fixed travel speeds (C = 28 kW-h, A/C on).

ProblemIns.		Flexible	Fixed travel speed (km/h)									
Type	ID	Speed	20		40		60		80		100	
		Total_cost	Total_cost	Dev.%	Total_cost	Dev.%	Total_cost	Dev.%	Total_cost	Dev.%	Total_cost	Dev.%
C1-type	C101	**540.82**	898.17	66.08	548.56	1.43	597.58	10.5	684.86	26.63	1328.81	145.7
	C102	**527.98**	530.78	0.53	545.9	3.39	597.57	13.18	684.5	29.65	823.78	56.02
	C103	**527.98**	530.78	0.53	545.9	3.39	597.57	13.18	684.5	29.65	823.78	56.02
	C104	**527.82**	530.78	0.56	545.9	3.43	597.57	13.21	684.37	29.66	823.78	56.07
	C105	**531.99**	896.51	68.52	548.56	3.11	597.58	12.33	684.74	28.71	1324.82	149.03
	C106	**535.71**	897.17	67.47	548.56	2.4	597.58	11.55	702.33	31.1	1327.82	147.86
	C107	**530.08**	530.14	0.01	548.56	3.49	597.58	12.73	684.74	29.18	823.33	55.32
	C108	**530.08**	530.14	0.01	548.56	3.49	597.58	12.73	684.64	29.16	823.33	55.32
	C109	**530.08**	530.14	0.01	548.56	3.49	597.58	12.73	684.74	29.18	808.49	52.52
	AVG	**531.39**	652.73	22.64	547.67	3.06	597.58	12.45	686.6	29.21	989.77	86.26

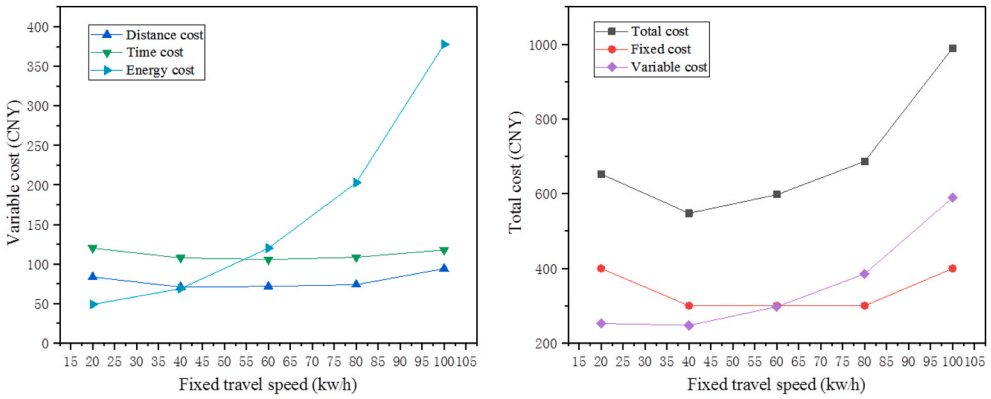
Note: Bold face indicates optimal values.

Table 13
Performances on C1-type instances with flexible and fixed travel speeds. (C = 28 kW-h, A/C on).

Fixed speed	Average base on 9C1-type instances								
	Total dis.	Total t.	Charge t.	Total cost	Fixed cost	Energy cost	Time cost	Dis. cost	CPU T.
(km/h)	(km)	(min)	(min)	(¥)	(¥)	(¥)	(c)	(¥)	(s)
$v_{ij} = 20$	69.67	1201.7	0	652.73	400	48.97	120.16	83.6	1.44
$v_{ij} = 40$	59.13	1077.89	0	547.67	300	68.93	107.79	70.96	1.67
$v_{ij} = 60$	59.71	1057.35	8.22	597.58	300	120.19	105.74	71.67	3.43
$v_{ij} = 80$	62.61	1085.49	46.11	686.6	300	202.93	108.55	74.02	5.03
$v_{ij} = 100$	78.52	1178	112.38	989.77	400	377.75	117.8	94.22	22.68
Flexible speed	59.13	1137.44	0	531.39	300	46.69	113.74	70.96	6.85

22.64, 3.06, 12.45, 29.21, and 86.26 % for the fixed speeds of 20, 40, 60, 80, and 100 km/h, respectively. Particularly at the fixed speed of 100 km/h, the cost increases are nearly 150 % for the C101, C105, and C106 instances. The average performances of the tested instances are summarized in Table 13 and illustrated in Fig. 14.

Fig. 14 (a) shows the trends of variable costs with the change in travel speed, including the distance cost, time cost, and energy cost. The time cost and distance cost show similar U-type curves that decrease first and then increase. However, the energy cost rises sharply from 48.97 to 377.75, when the fixed speed was increased from 20 km/h to 100 km/h. This is because the high travel speed of 100 km/h also has a very high ECR, such that EVs must detour more frequently to visit the CSs. This



(a) Variable cost variation trends (b) Total cost variation trends

Fig. 14. Trends of costs with respect to different fixed travel speeds.

also prompt that the tested EV has a quite worse energy efficiency when the speed is high, and the driving range could be shortened considerably in this case. Fig. 14 (b) shows the cost trends of variable cost, fixed cost, and total cost (variable cost plus fixed cost). The variable cost undergoes a steady increase with speed, while the fixed cost shows a U-type curve. This is because the fixed cost is directly related to the number of vehicles used. Fig. 14 (b) also hints that the economic speed for the tested EVs is around 40 km/h, much lower than those of most known fossil-fuel powered vehicle models.

7. Conclusions

This study reviewed the development of the EVRP. Then, by considering the dynamic influence of speed and payload changes on the maximum remaining range of EVs, an improved ECR function was proposed by summing up all types of consumptions arising from aerodynamic, tire, drivetrain, ancillary, and kinetic/potential losses. The nonlinear ECR function was linearized by bounding travel time to travel speed using a set of tangential lines. Meanwhile, by factoring in the electricity/energy consumption model and nonlinear charge function, an EVRPTW model with a new approach to routing formulation with the nearest CS visit was developed.

Listed below are a few managerial insights for practical logistics using EVs and conclusions of the study based on the computational experiments on Solomon's instances:

(1) Minimize time with air conditioners on. Comparative experiments showed the total consumed energy with the A/C turned off on average was −13.31 % lower than those with the A/C turned on. Sometimes, the reduction rates were nearly −20 %. Accordingly, the minimum total cost (¥) with the A/C turned off is lower than that with the A/C turned on.

(2) Choose a larger battery capacity for the EV. The solutions produced with a battery capacity of 32 kW-h consistently incurred lower total costs than those obtained with other battery capacities, with average cost deviations of 21.23, 9.73, 1.70, and 0.19 % with respect to the battery capacities of 16, 20, and 24 kW-h, respectively. In general, all variable costs decreased steadily as the battery capacity increased from 16 to 32 kW-h, and remain stable thereafter. However, in realities the fixed cost of an EV may also get increased for a larger battery capacity, so there may exist an optimal capacity that is a trade-off of balancing the fixed and variable costs.

(3) Drive at a flexible travel speed on each route rather than at a fixed speed. The solutions obtained at different fixed travel speeds consistently incurred higher total costs than those obtained at flexible travel speeds. The average rate of cost increase can be 22.64, 3.06, 12.45, 29.21, and 86.26 % for the fixed speeds of 20, 40, 60, 80, and 100 km/h, respectively.

The EVRPTW model, which represents real logistics situations, helped optimize the total cost consisting of time, distance, energy, and fixed costs, by having the EV drive at a flexible travel speed and charging time under different SOC conditions. Future research should focus on developing more efficient heuristic algorithms to solve large-scale problems.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors thank JB Straubel for his/her ECR model provided on the website which facilitates this study. This study was partly supported by the National Natural Science Foundation of China under Grant Nos. 71871003 and 719710009.

References

[1] Mccarthy M, Best M, Betts R. Climate change in cities due to global warming and urban effects. Geophys Res Lett 2010;37(9).
[2] Sims R, Schaeffer R, et al. Climate change 2014: mitigation of climate change. Contribution of working group III to the fifth assessment report of the intergovernmental panel on climate change. Cambridge, United Kingdom and New York, USA: Cambridge University Press; 2014.
[3] Zuo X, Xiao Y, You M, Kaku I, Xu Y. A new formulation of the electric vehicle routing problem with time windows considering concave nonlinear charging function. J Clean Prod 2019;236:117687.
[4] Soleimani H, Chaharlang Y, Ghaderi H. Collection and distribution of returned-remanufactured products in a vehicle routing problem with pickup and delivery considering sustainable and green criteria. J Clean Prod 2018;172:960–70.
[5] Desaulniers G, Errico F, Irnich S, Schneider M. Exact algorithms for electric vehicle-routing problems with time windows. Oper Res 2016;64(6):1388–405.
[6] Li C, Cao Y, Zhang M, et al. Hidden benefits of electric vehicles for addressing climate change. Sci Rep 2015;5(1):9213.
[7] Xiao Y, Zuo X, Kaku I, Zhou S, Pan X. Development of energy consumption optimization model for the electric vehicle routing problem with time windows. J Clean Prod 2019;225:647–63.
[8] Qian K, Zhou C, Allan M, et al. Modeling of load demand due to EV battery charging in distribution systems. IEEE Trans Power Syst 2011;26(2):802–10.
[9] Mashayekh Y, Jaramillo P, Samaras C, et al. Potentials for sustainable transportation in cities to alleviate climate change impacts. Environ Sci Technol 2012;46(5):2529–37.
[10] Eberle U, Helmolt R. Sustainable transportation based on electric vehicle concepts: a brief overview. Energy Environ Sci 2010;3(6):689.
[11] Offer GJ, Howey D, Contestabile M, et al. Comparative analysis of battery electric, hydrogen fuel cell and hybrid vehicles in a future sustainable road transport system. Energy Pol 2010;38(1):24–9.

[12] Pucher J, Buehler R. Making cycling irresistible: lessons from The Netherlands, Denmark and Germany. Transport Rev 2008;28(4):495–528.
[13] International Energy Agency (IEA). Global EV outlook 2018. 2018. https://www.indiaenvironmentportal.org.in/files/file/globalevoutlook2018.pdf. 25 July 2018.
[14] Erdoğan S, Miller-Hooks E. A green vehicle routing problem. Transport Res E Logist Transport Rev 2012;48(1):100–14.
[15] Goeke D, Schneider M. Routing a mixed fleet of electric and conventional vehicles. Eur J Oper Res 2015;245(1):81–99.
[16] Felipe A, Ortuño MT, Righini G, Tirado G. A heuristic approach for the green vehicle routing problem with multiple technologies and partial recharges. Transport Res E Logist Transport Rev 2014;71:111–28.
[17] Montoya A, Gueret C, Mendoza JE, Villegas JG. The electric vehicle routing problem with nonlinear charging function. Transp Res Part B Methodol 2017;103:87–110.
[18] Conrad R, Figliozzi M. The recharging vehicle routing problem. In: Proceedings of the 2011 industrial engineering research conference; 2011. p. 111–8.
[19] Bard J, Huang L, Dror M, Jaillet P. A branch and cut algorithm for the VRP with satellite facilities. IIE Trans 1998;30(9):821–34.
[20] Yang J, Sun H. Battery swap station location-routing problem with capacitated electric vehicles. Comput Oper Res 2015;55:217–32.
[21] Hiermann G, Puchinger J, Ropke S, Hartl R. The electric fleet size and mix vehicle routing problem with time windows and recharging stations. Eur J Oper Res 2016;252(3):995–1018.
[22] Iqbal S, Rahman M. Vehicle routing problems with soft time windows. In: International conference on electrical & computer engineering. IEEE; 2013.
[23] Koç Ç, Karaoglan I. The green vehicle routing problem: a heuristic based exact solution approach. Appl Soft Comput 2016;39:154–64.
[24] Leggieri V, Haouari M. A practical solution approach for the green vehicle routing problem. Transport Res E Logist Transport Rev 2017;104:97–112.
[25] Penna PHV, Afsar HM, Prins C, et al. A hybrid iterative local search algorithm for the electric fleet size and mix vehicle routing problem with time windows and recharging stations. IFAC-PapersOnLine 2016;49(12):955–60.
[26] Pelletier S, Jabali O, Laporte G, Veneroni M. Battery degradation and behaviour for electric vehicles: review and numerical analyses of several models. Transp Res Part B Methodol 2017;103:158–87.
[27] Hõimoja H, Rufer A, Dziechciaruk G, Vezzini A. An ultrafast EV charging station demonstrator. In: 2012 international symposium on power electronics. Electrical drives, automation and motion (SPEEDAM). IEEE; 2012. p. 1390–5.
[28] Adler JD, Mirchandani PB. Online routing and battery reservations for electric vehicles with swappable batteries. Transp Res Part B Methodol 2014;70:285–302.
[29] Montoya A, Guéret C, Mendoza JE, Villegas JG. A multi-space sampling heuristic for the green vehicle routing problem. Transport Res C Emerg Technol 2015;70:113–28.
[30] Hof J, Schneider M, Goeke D. Solving the battery swap station location-routing problem with capacitated electric vehicles using an AVNS algorithm for vehicle-routing problems with intermediate stops. Transp Res Part B Methodol 2017;97:102–12.
[31] Schneider M, Stenger A, Goeke D. The electric vehicle-routing problem with time windows and recharging stations. Transport Sci 2014;48(4):500–20.
[32] Schneider M, Stenger A, Hof J. An adaptive VNS algorithm for vehicle routing problems with intermediate stops. Spectrum 2015;37(2):353–87.
[33] Lin J, Zhou W, Wolfson O. Electric vehicle routing problem. Trans Res Procedia 2016;12:508–21.
[34] Szeto WY, Cheng Y. Artificial bee colony approach to solving the electric vehicle routing problem. J Eastern Asia Soc Tran Stud 2017;12:975–90.
[35] Sassi O, Cherif WR, Oulamara A. Vehicle routing problem with mixed fleet of conventional and heterogeneous electric vehicles and time dependent charging costs. Int J Math Comput Sci 2015;9(3):163–73.
[36] Bülent C, Merve K. A matheuristic approach for solving the electric vehicle routing problem with time windows and fast recharges. Column Gen 2016:67–98.
[37] Oriol T, Marti R, Eric J. Transport energy consumption in mountainous roads. A comparative case study for internal combustion engines and electric vehicles in Andorra. Transport Res Transport Environ 2015;34:16–26.
[38] Bruglieri M, Pezzella F, Pisacane O, Suraci S. A variable neighborhood search branching for the electric vehicle routing problem with time windows. Electron Notes Discrete Math 2015;47:221–8.
[39] Schiffer M, Walther G. The electric location routing problem with time windows and partial recharging. Eur J Oper Res 2017;260(3):995–1013.
[40] Vermaa A, Lamsalb K, Keoughb S. Electric vehicle routing problem with time windows, recharging stations and battery swapping stations. Kansas: ESU; 2015.
[41] Keskin M, Çatay B. Partial recharge strategies for the electric vehicle routing problem with time windows. Transport Res C Emerg Technol 2016;65:111–27.
[42] Worley O, Klabjan D. Optimization of battery charging and purchasing at electric vehicle battery swap stations. In: 2011 IEEE vehicle power and propulsion conference, chicago, IL; 2011. p. 1–4.
[43] Mak H, Rong Y, Shen Z. Infrastructure planning for electric vehicles with battery swapping. Manag Sci 2013;59(7):1557–75.
[44] Avci B, Girotra K, Netessine S. Electric vehicles with a battery switching station: adoption and environmental impact. Manag Sci 2014;61(4):772–94.
[45] Yang H, Yang S, Xu Y, et al. Electric vehicle route optimization considering time-of-use electricity price by learnable partheno-genetic algorithm. IEEE Trans Smart Grid 2015;6(2):657–66.
[46] Chen J, Qi M, Miao L. The electric vehicle routing problem with time windows and battery swapping stations. In: 2016 IEEE international conference on industrial engineering and engineering management (IEEM); 2016.
[47] Marmaras C, Xydas E, Cipcigan L. Simulation of electric vehicle driver behaviour in road transport and electric power networks. Transport Res C Emerg Technol 2017;80:239–56.
[48] Liu K, Yamamoto T, Morikawa T. Impact of road gradient on energy consumption of electric vehicles. Transport Res Transport Environ 2017;54:74–81.
[49] Javani N, Dincer I, Naterer GF. Thermodynamic analysis of waste heat recovery for cooling systems in hybrid and electric vehicles. Energy 2012;46(1):109–16.
[50] Fiori C, Ahn K, Rakha HA. Power-based electric vehicle energy consumption model: model development and validation. Appl Energy 2016;168(15):257–68.
[51] Yuan X, Zhang C, Hong G, Huang X, Li L. Method for evaluating the realworld driving energy consumptions of electric vehicles. Energy 2017;141:1955–68.
[52] Barco J, Guerra A, Muñoz L, et al. Optimal routing and scheduling of charge for electric vehicles: a case study. Math Probl Eng 2017;16.
[53] Fetene M, Kaplan S, Mabit L, Jensen F, Prato G. Harnessing big data for estimating the energy consumption and driving range of electric vehicles. Transport Res Transport Environ 2017;54:1–11.
[54] Ding N, Batta R, Kwon C, et al. Conflict-free electric vehicle routing problem with capacitated charging stations and partial recharge. Buffalo: SUNY; 2015.
[55] Froger A, Mendoza JE, Jabali O, et al. A matheuristic for the electric vehicle routing problem with capacitated charging stations. HAL ID; 2017. hal-01559524.
[56] Kchaou-Boujelben M, Gicquel C. Locating electric vehicle charging stations under uncertain battery energy status and power consumption. Comput Ind Eng 2020;149:106752.
[57] Soysal M, Çimen M, Belbağ S. Pickup and delivery with electric vehicles under stochastic battery depletion. Comput Ind Eng 2020;146:106512.
[58] Golden B, Assad A, Levy L, et al. The fleet size and mix vehicle routing problem. Comput Oper Res 1984;11(1):49–66.
[59] Booth K, Beck J. A constraint programming approach to electric vehicle routing with time windows. In: Rousseau LM, Stergiou K, editors. Integration of constraint programming, artificial intelligence, and operations research. CPAIOR 2019. Lecture notes in computer science, vol. 11494. Cham: Springer; 2019.
[60] Juan AA, Goentzel J, Bekta T. Routing fleets with multiple driving ranges: is it possible to use greener fleet configurations? Appl Soft Comput 2014;21:84–94.
[61] Rezgui D, Siala CJ, Aggoune-Mtalaa W, Bouziri H. Application of a variable neighborhood search algorithm to a fleet size and mix vehicle routing problem with electric modular vehicles. Comput Ind Eng 2019;130:537–50.
[62] Gonçalves F, Cardoso SR, Relvas S, et al. Optimization of a distribution network using electric vehicles: AVRP problem. In: Proc of the IO2011-15 congressoda associação portuguesa de Investigação operacional coimbra: 18–20; 2011.
[63] Montazeri-Gh M, Poursamad A, Ghalichi B. Application of genetic algorithm for optimization of control strategy in parallel hybrid electric vehicles. J Franklin Inst 2006;343(4–5):420–35.
[64] Zuo X, Zhu C, Huang C, Xiao Y. Using AMPL/CPLEX to model and solve the electric vehicle routing problem (EVRP) with heterogeneous mixed fleet. In: 2017 29th Chinese control and decision conference (CCDC); 2017.
[65] Van Duin J, Tavasszy L, Quak H. Towards electric-urban freight: first promising steps in the electric vehicle revolution. Eur Trans 2013;54:1–9.
[66] Andelmin J, Bartolini E. An exact algorithm for the green vehicle routing problem. Transport Sci 2017;51(4):1288–303.
[67] Lebeau P, De Cauwer C, Van Mierlo J, et al. Conventional, hybrid, or electric vehicles: which technology for an urban distribution centre? Sci World J 2015:1–11.
[68] Ángel Felipe, et al. A heuristic approach for the green vehicle routing problem with multiple technologies and partial recharges. Transport Res E Logist Transport Rev 2014;71:111–28.
[69] Omidvar A, Tavakkoli-Moghaddam R. Sustainable vehicle routing: strategies for congestion management and refueling scheduling. In: 2012 IEEE int energy conf. And exhibit. Florence: ENERGYCON); 2012. p. 1089–94.
[70] Preis H, Frank S, Nachtigall K. Energy-optimized routing of electric vehicles in urban delivery systems. In: Operations research proceedings 2012; 2013. p. 583–8.
[71] Bruglieri M, Mancini S, Pezzella F, et al. A three-phase matheuristic for the time-effective electric vehicle routing problem with partial recharges. Electron Notes Discrete Math 2017;58:95–102.
[72] Bräysy O, Dullaert W, Hasle G, et al. An effective multirestart deterministic annealing metaheuristic for the fleet size and mix vehicle-routing problem with time windows. Transport Sci 2008;42(3):371–86.
[73] Wang H, Shen J. Heuristic approaches for solving transit vehicle scheduling problem with route and fueling time constraints. Appl Math Comput 2007;190(2):1237–49.
[74] Boschetti MA, Maniezzo V, Roffilli M, et al. Matheuristics: optimization, simulation and control. In: Blesa MJ, Blum C, Di Gaspero L, Roli A, Sampels M, Schaerf A, editors. Hybrid metaheuristics. HM 2009. Lecture notes in computer science, vol. 5818. Berlin, Heidelberg: Springer; 2009.
[75] Doerner KF, Schmid V. Survey: matheuristics for rich vehicle routing problems. Hybrid Metaheuristics 2010;6373:206–21.
[76] Archetti C, Speranza MG. A survey on matheuristics for routing problems. EURO J Comput Opt 2014;2(4):223–46.
[77] Wang H, Liu Y, Fu H, Li G. Estimation of state of charge of batteries for electric vehicles. Int J Contr Auto 2013;6(2):185–94.
[78] Zhang X, Yao J, Liao Z, Li J. The electric vehicle routing problem with soft time windows and recharging stations in the reverse logistics. In: Xu J, Cooke F, Gen M, Ahmed S, editors. Proc. Of the twelfth int. Conf. On management science and engineering management. ICMSEM 2018. Lecture notes on multidisciplinary industrial engineering. Cham: Springer; 2019.
[79] Solomon Marius M. Algorithms for the vehicle routing and scheduling problems with time window constraints. Oper Res 1987;35(2):254–65.